香港科技大学(广州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

UCUG 1903 Signals and Information Technology

2026-27 Fall Term

Lecture 3 Spectrum and Filters

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Dr. Shanshi HUANG

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Instructor Information

- **Instructor**: Dr. Shanshi HUANG
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- **Lecture Time & Location**: Wednesday 16:30-18:20, W4-101
- **Tutorial Time & Location**: Thursday 12:00-12:50, E1-201

(Please bring your laptop and pre-install MATLAB using the provided guide!)

- **Office Hour**: Monday 14:00-15:00 W2 607
- **Teaching Assistants:** Mr. Xunqin Lai
- **Email:** xlai701@connect.hkust-gz.edu.cn

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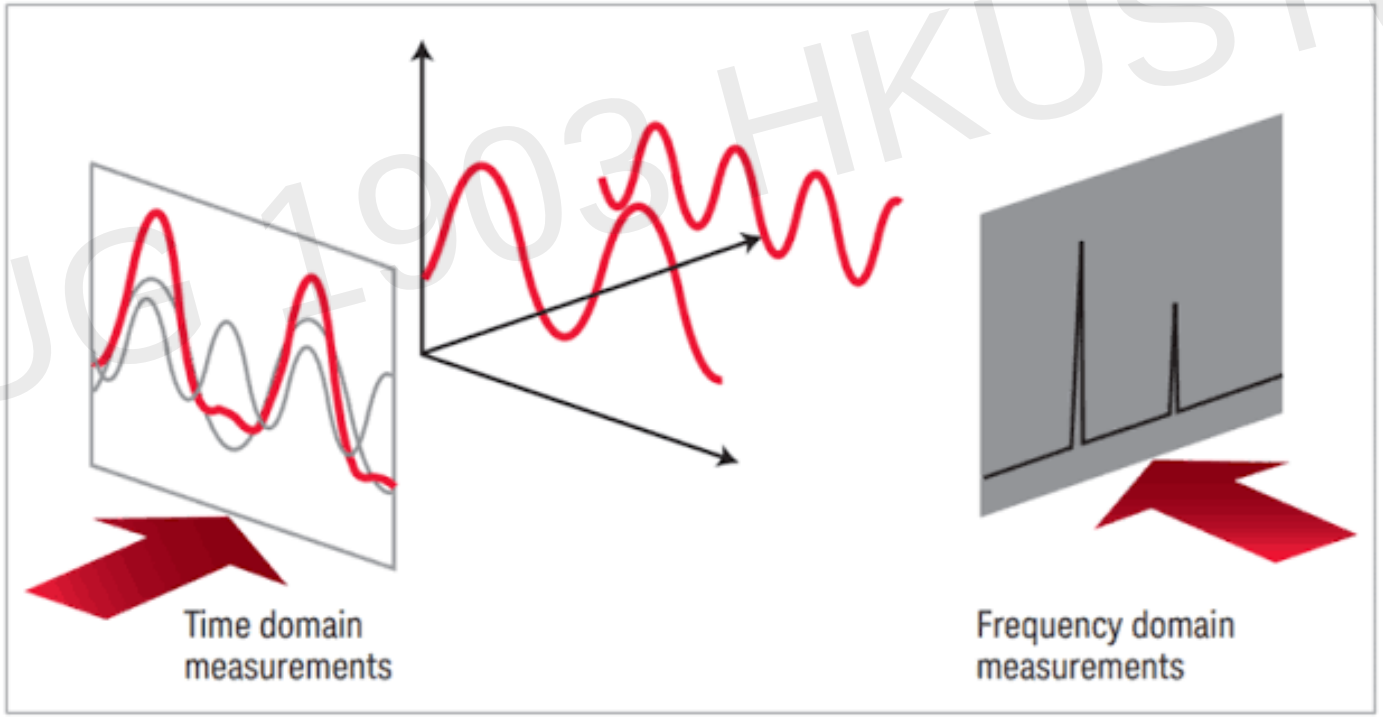
Source: ChatGPT

Spectrum

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Representing Signals in Different Domain

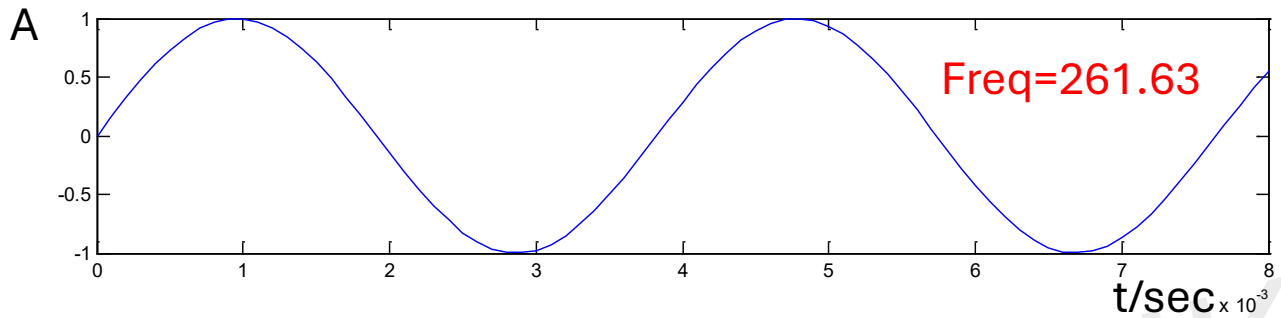
- From Lecture 2, we know that "Any periodic patterns can be created by adding up different amounts of harmonics!
- This gives us *two* equivalent ways to represent signals!
  - Time-domain representation: plot the entire periodic amplitude variations over time
  - Frequency-domain representation: plot the amplitudes of the **sine waves** at different frequencies



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Representation of Sine Wave

Time Domain Representation: Waveform



Axis: time

Frequency Domain Representation: Spectrum



Do 

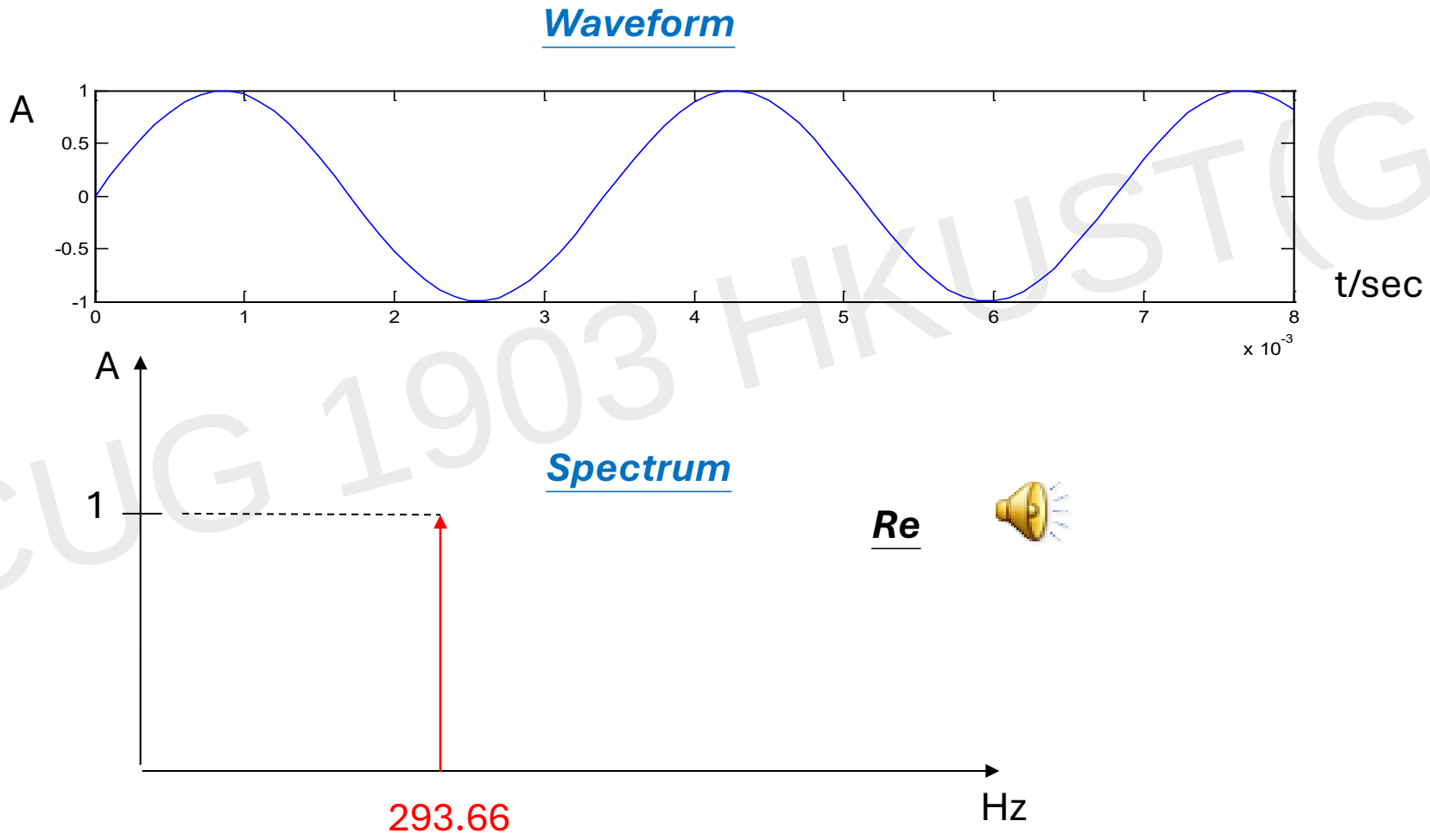
Axis: frequency

- Our perception of sound is closely related to its **frequency components**.
- A sine wave at **261.63 Hz** with amplitude **1** can be represented by a single arrow at **261.63 Hz**, with a height of **1**.
- This makes the frequency information **more direct and compact** than in the time-domain waveform.

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Waveform vs. Spectrum

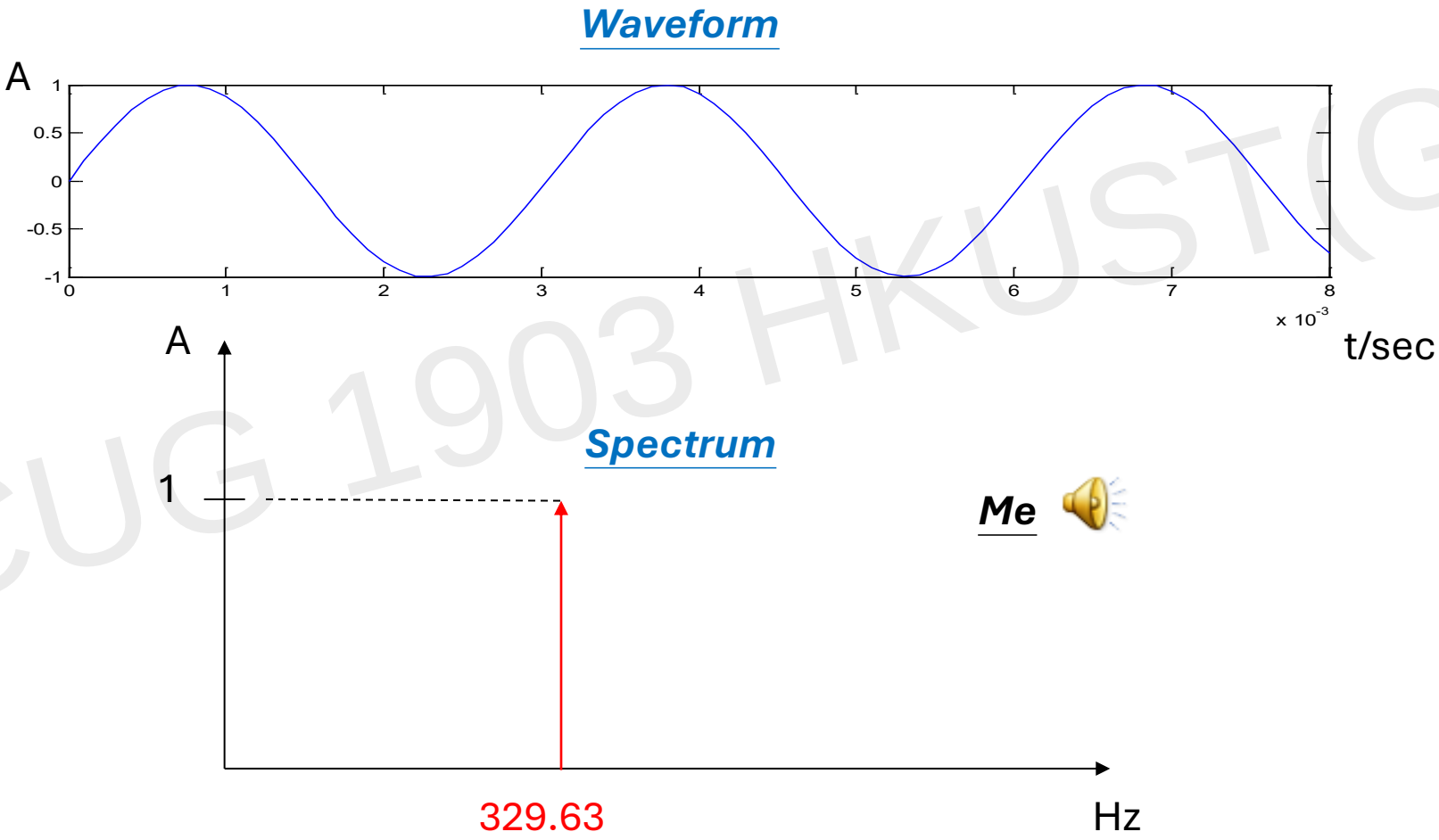
- If we have a sine wave at 293.66 Hz. We draw an arrow at 293.66 Hz



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Waveform vs. Spectrum

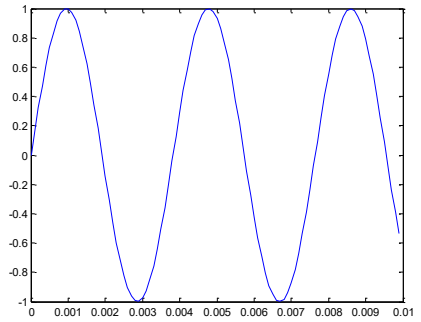
- 329.63 Hz



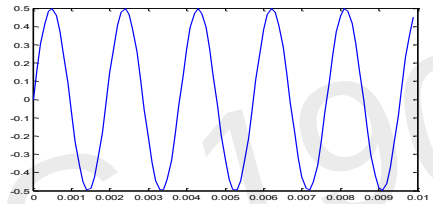
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Representation of Complex Signal

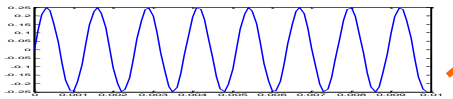
1st harmonic (Fundamental frequency)



2nd harmonic

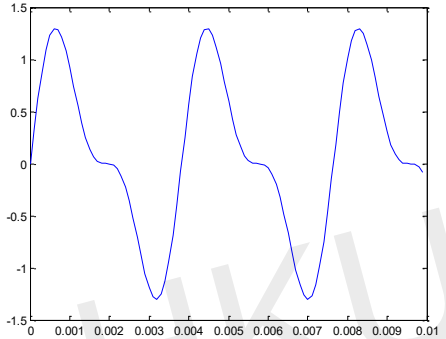


3rd harmonic



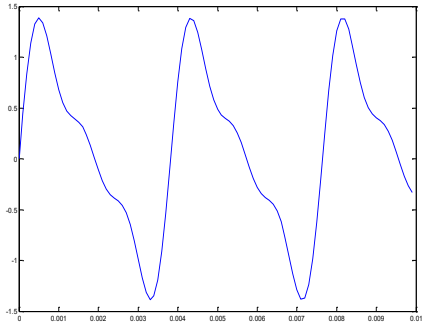
Signal $x_1(t)$

1st + 2nd harmonics

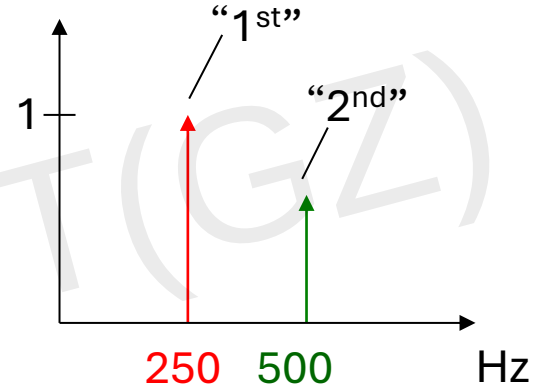


Signal $x_2(t)$

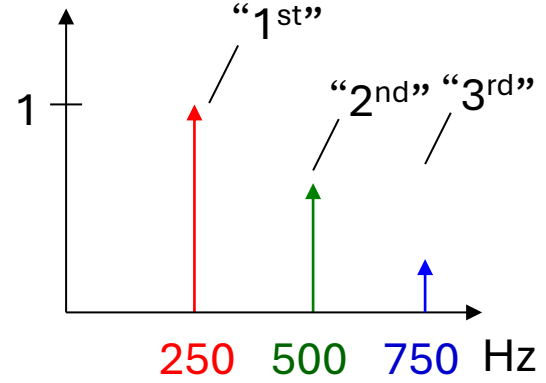
1st +2nd+3rd harmonics



Spectrum $X_1(f)$



Spectrum $X_2(f)$

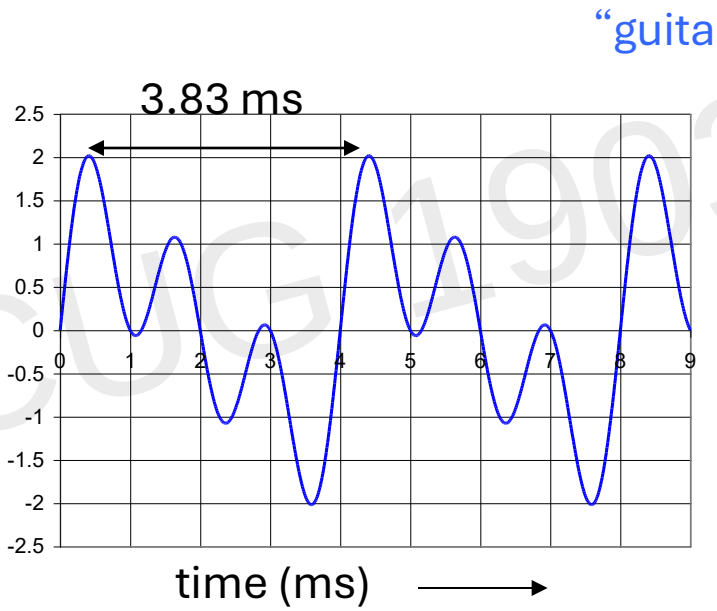


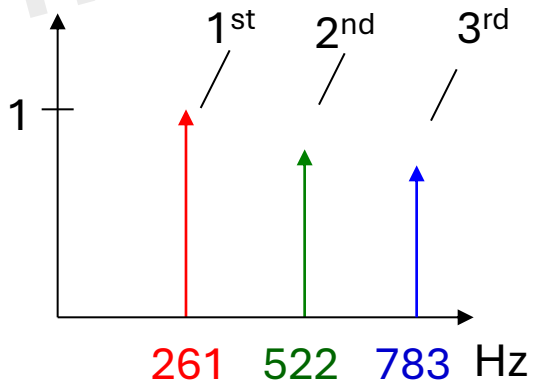
We can readily explain how the two signals are different by referring to their ***spectra***!

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Spectrum as Energy at Different Frequencies

- The spectrum represents the amplitude, or the amount of energy, in the signal at different frequencies.
- For example, the spectrum below tells us that when we pluck the “C” note on the guitar, the energy is concentrated at the 1st, 2nd and 3rd harmonics.

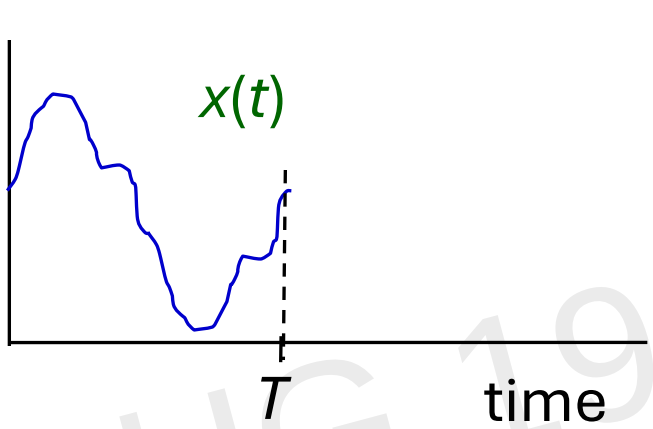




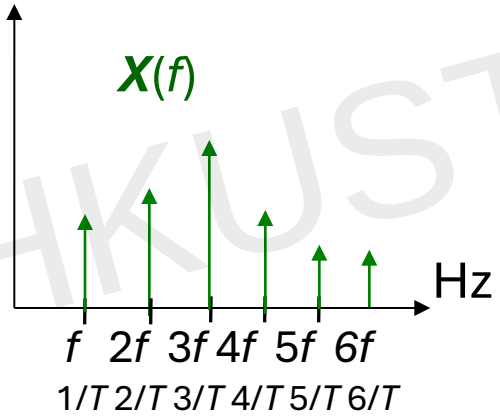
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Any Signal as A Sum of Sine Waves

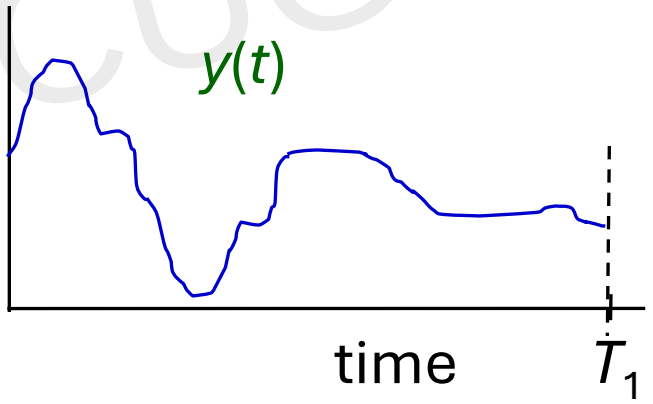
- We can go one step further – we can simply take any chunk of signal $x(t)$ over an interval T, and we can treat the signal as if it is periodic with period T, and we can find the set of harmonic sine waves that will sum up to this signal.



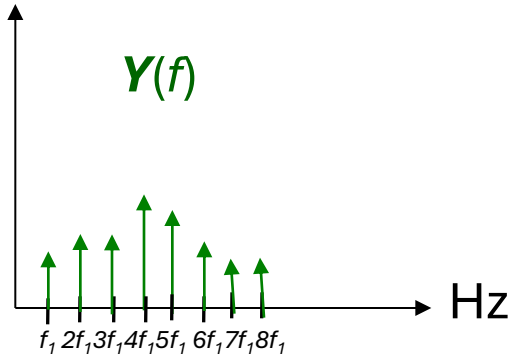
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Amplitude of sine waves at different harmonic frequencies



=

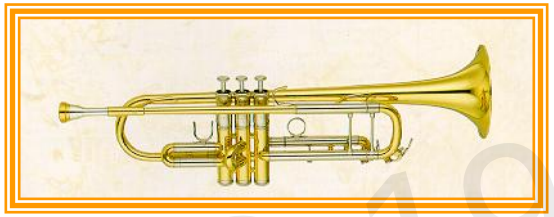


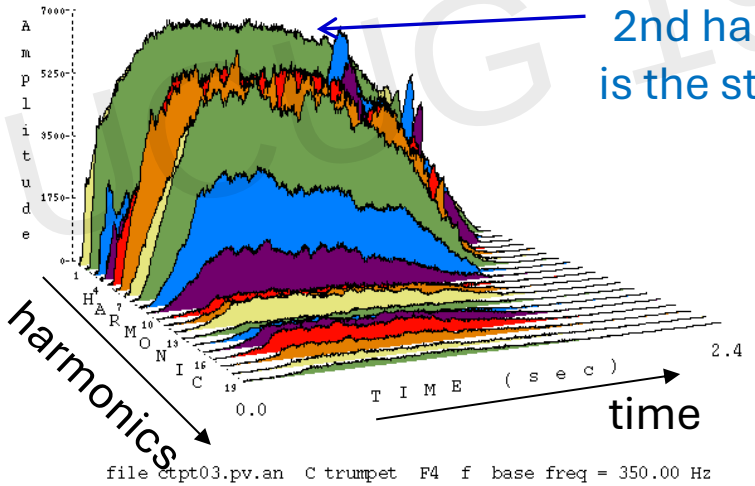
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Spectrogram

- A signal may have different spectra at different times.
- Even for the same note, the amplitudes of its harmonics may vary with time.
- A **spectrogram** shows how the spectrum of a signal changes over time.

Trumpet

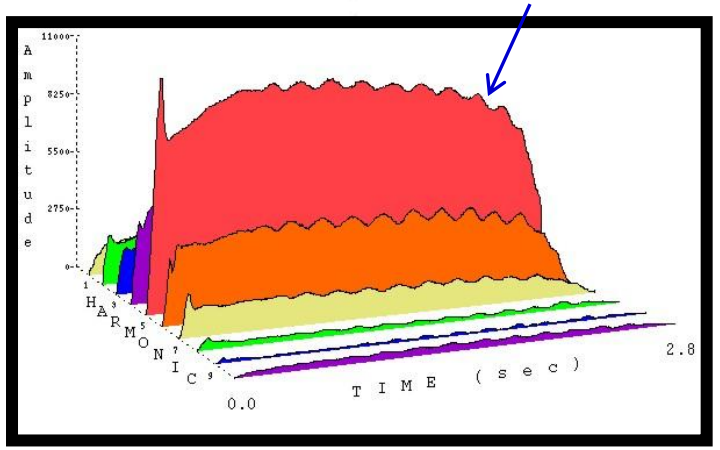




Clarinet



Most energy in 5th harmonic



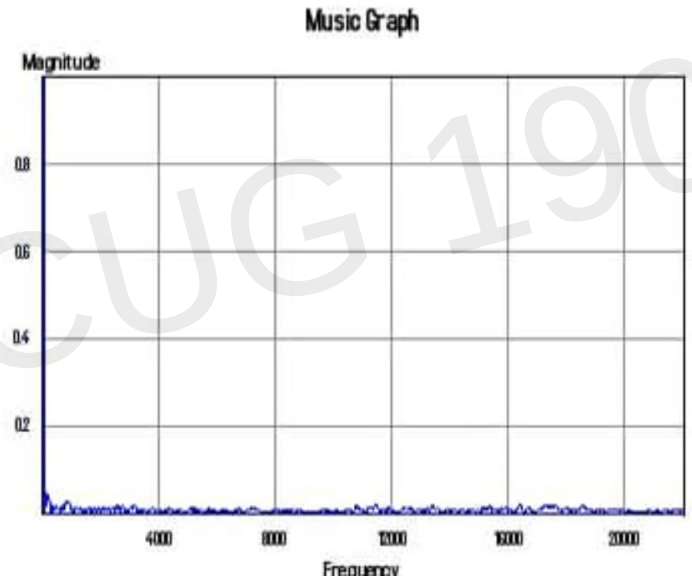
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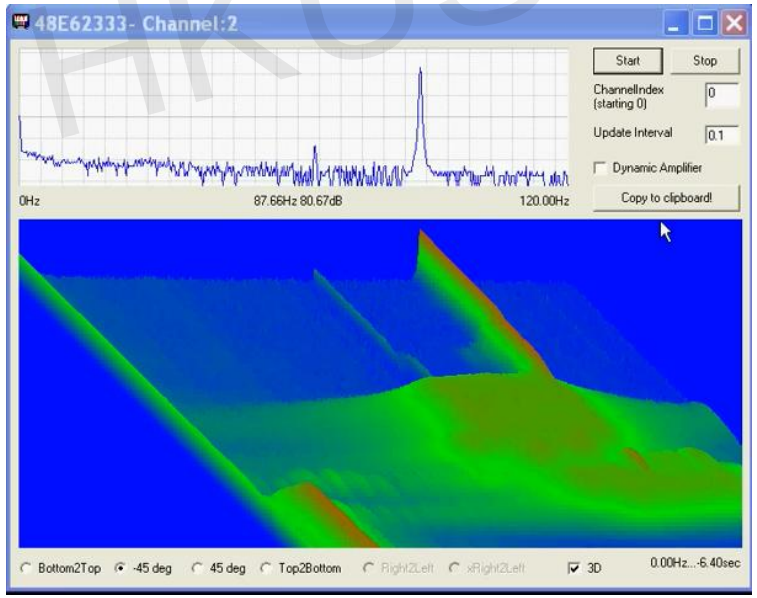
Spectrum Analyzer of A Real-Time Signal in Action

- In the Internet demo below, we can see a spectrum analyzer in action analyzing a real time music. The spectrum analyzer cuts up the music in time into "chunks" and generates the spectrum of each chunk.

- A real-time analysis of a song. This spectrum analyzer output features frequency on *X* (horizontal), magnitude on *Y* (vertical), and moves through time in sequence with the song

- We can see a similar demo here. The result is a spectrogram in the form of a waterfall.





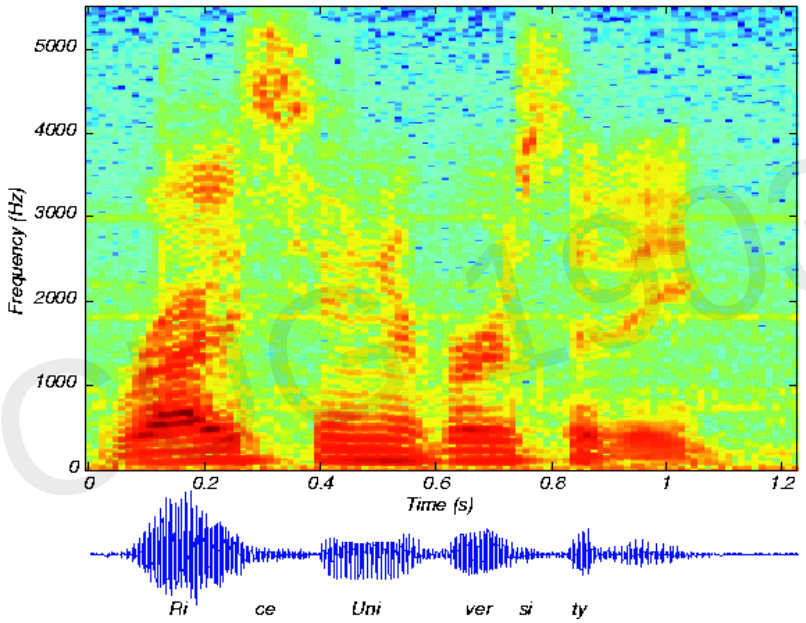
http://en.wikipedia.org/wiki/File:My_Songo_Real_Time_Analysis.ogg
http://www.ultimaserial.com/UltimaWaterfall.html

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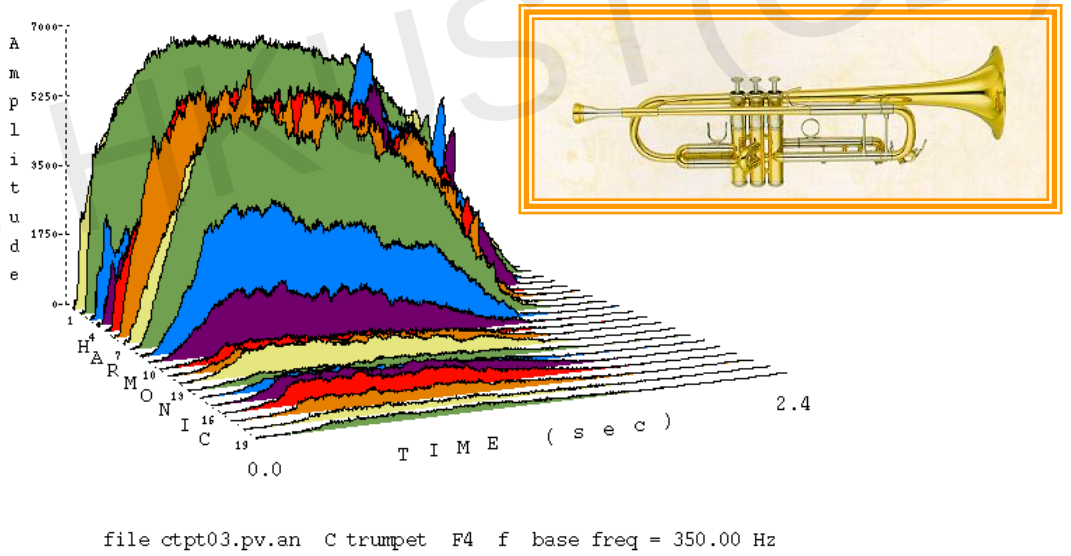
Showing Spectrogram in 2-D or 3-D

- Without use of animation, we can also show a *spectrogram* as a 2-D or 3-D plot below. In the 2-D plot, we use color to indicate the amplitude of individual frequencies at different times.

A Speech Spectrogram in 2D Plot



A Trumpet Spectrogram in 3D Plot

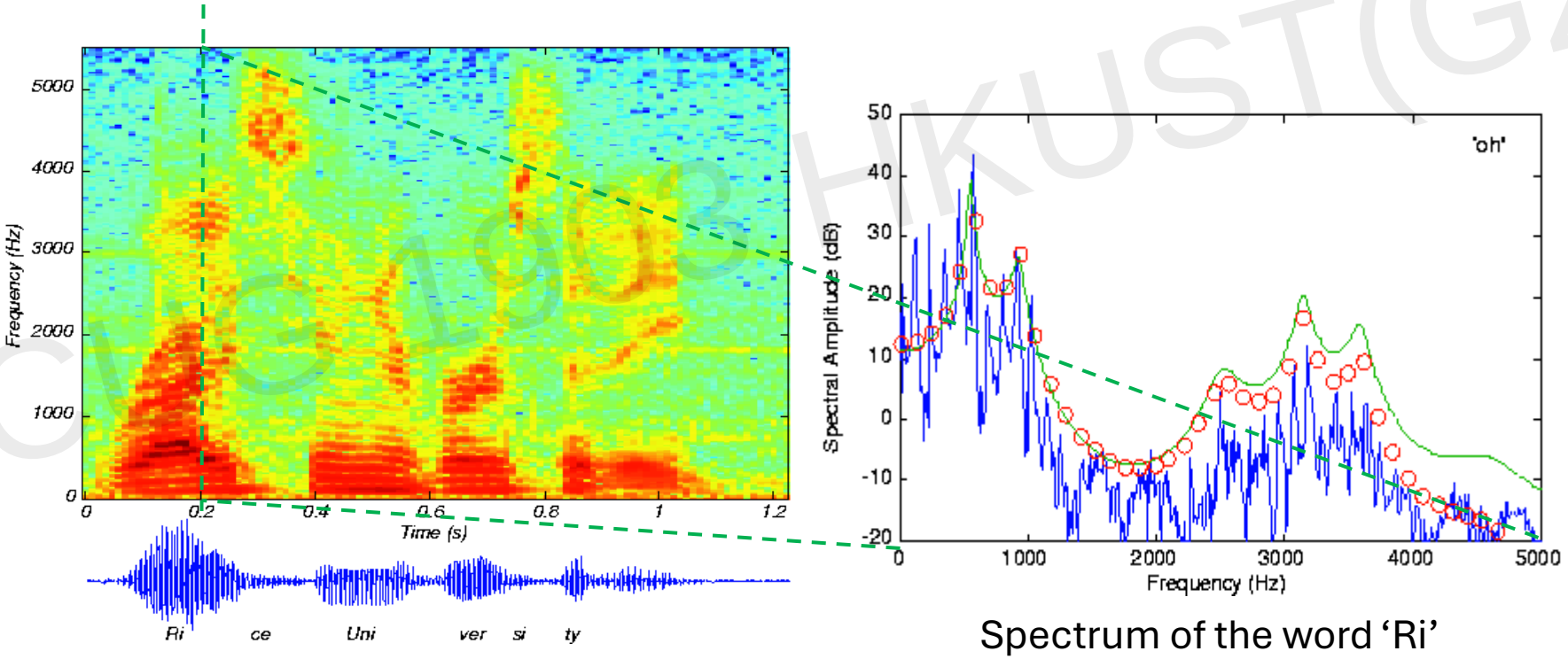


- Speech Spectrum as a function of time. Red means large amplitude and blue means small amplitude.

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Spectrogram and Spectrum

- A **spectrogram** can be viewed as a sequence of spectra arranged over time.
- A **spectrum** can be viewed as a cross-section of the spectrogram at a particular time.



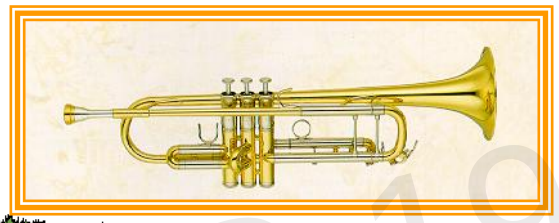
Spectrum of the word ‘Ri’

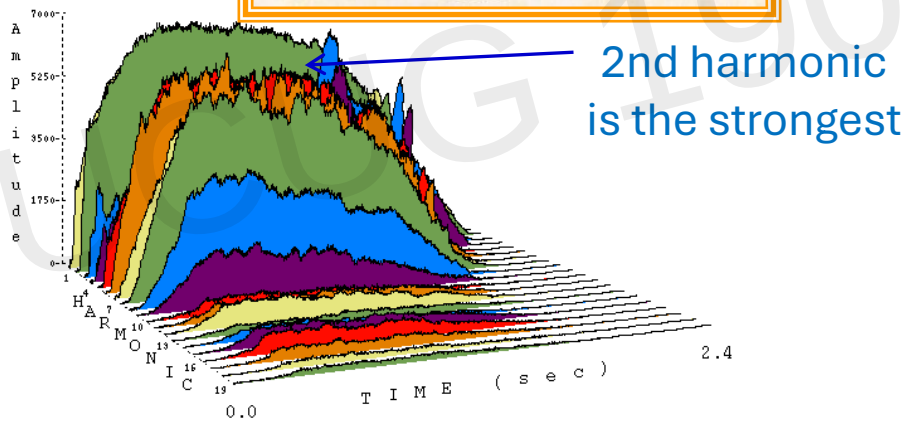
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Why Spectrogram?

- Compared with a spectrum, a spectrogram shows both harmonic structure and temporal evolution.
- This can provide a more complete description of a signal, such as sound timbre.

Trumpet





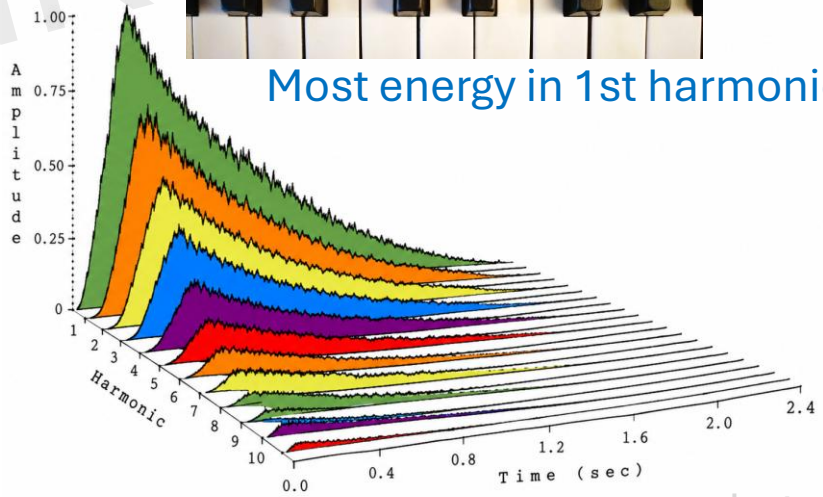
2nd harmonic
is the strongest

file ctpt03.pv.an C trumpet F4 f base freq = 350.00 Hz

Piano



Most energy in 1st harmonic



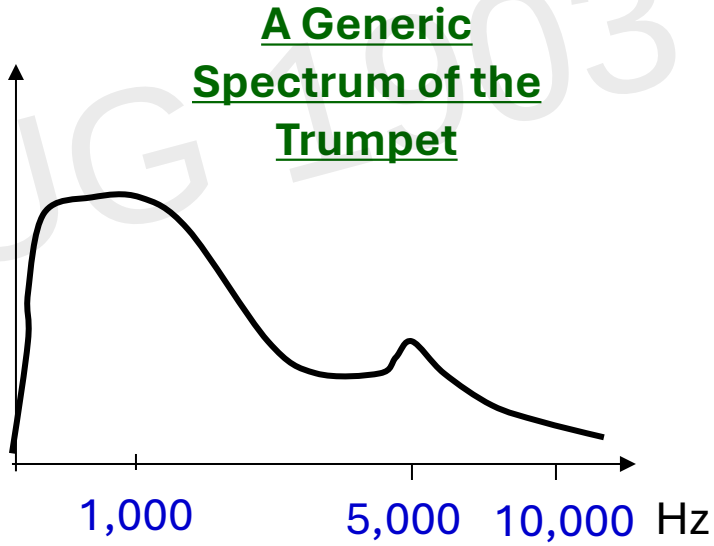
piano note C4 base freq ≈ 261.6 Hz

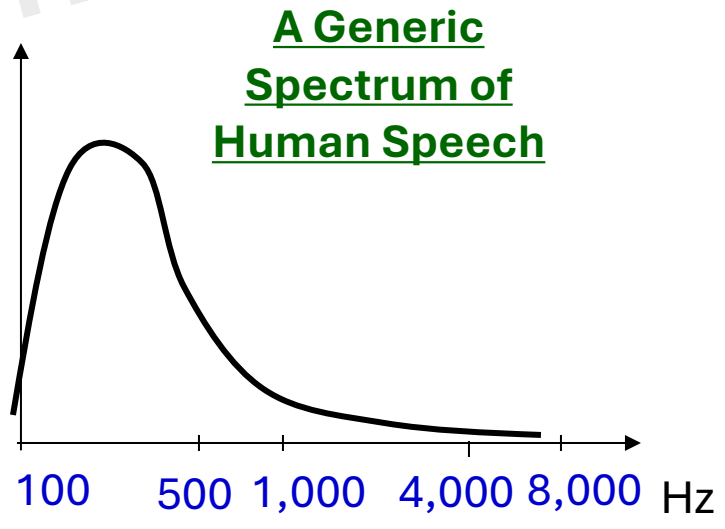
chatgpt

Average Spectrum

- An **average spectrum** shows the **average energy at each frequency over time**.
- By averaging the spectra from many short time intervals, short-term variations are smoothed out, giving a more stable and compact view of the signal's frequency content.
- The average spectrum can therefore reveal the **overall spectral characteristics** of the signal source.

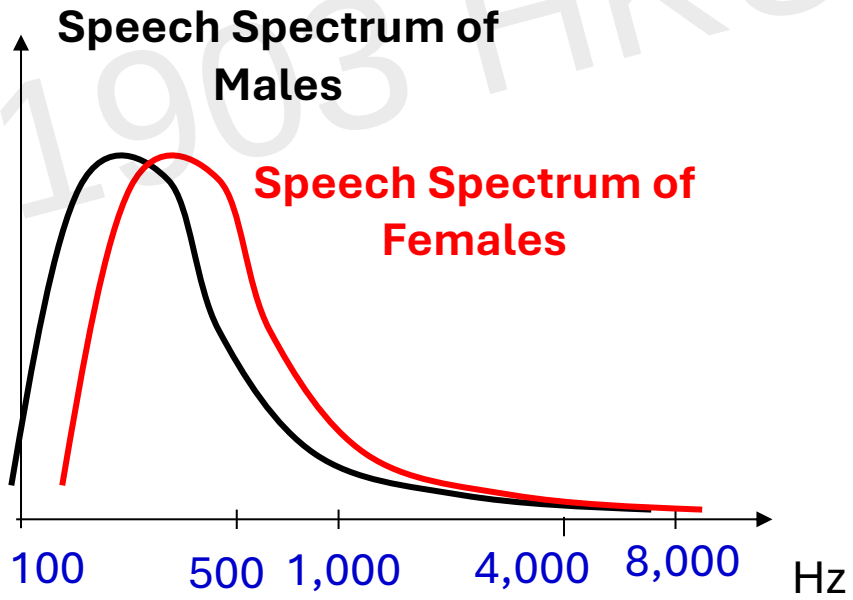
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Property of Signals from Their Spectra

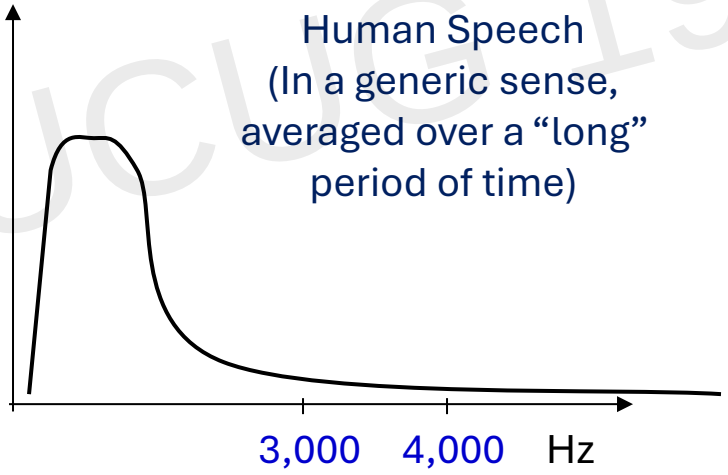
- The (average) spectrum allows us to discuss the properties of the signal in a generic way. For example:
  - Human speech spectrum has negligible energy at frequency > 4,000 Hz.
  - Female speech has higher frequencies than male speech.
  - Also, there is no energy at very low frequency.

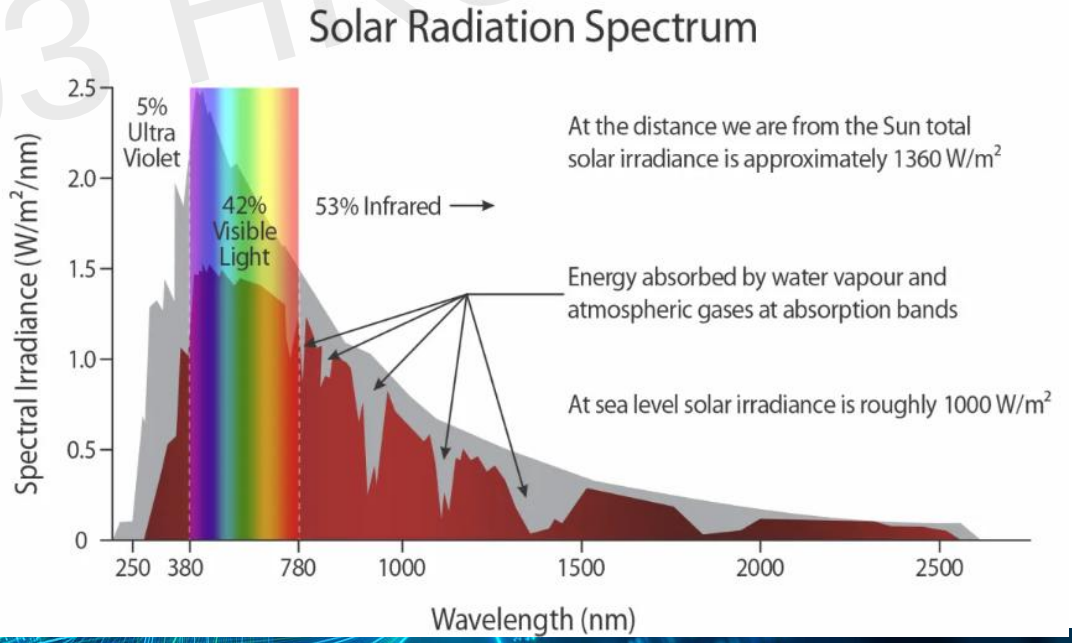


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From sound to electromagnetic wave

- For the remainder of this course, we will forget about the spectrogram and use ***spectrum*** to specify the distribution of signal at different frequencies in a generic sense.
- Not only that the concept of spectrum is applied to acoustic signals, but it is also widely applied to the understanding of *electromagnetic wave*.
- Radiation from the sun, for example, is made up of electromagnetic waves at various frequencies.

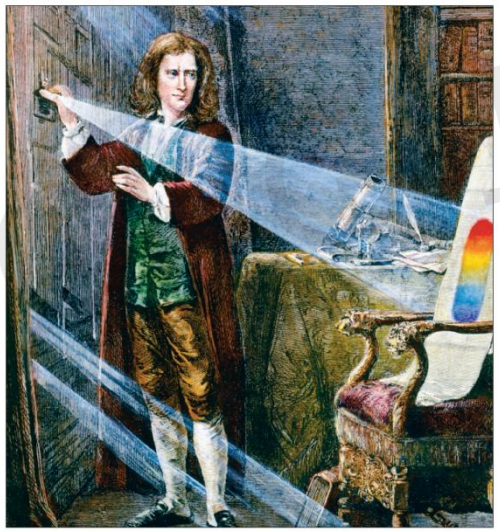


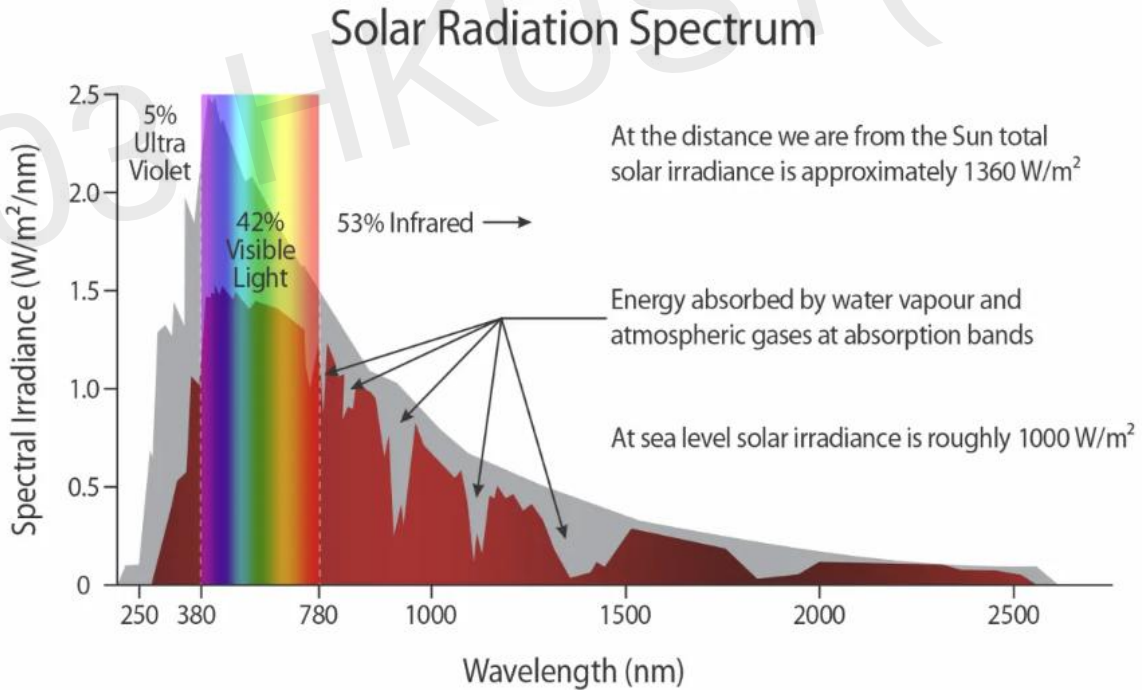


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Newton and The Spectrum of Light

- Newton, by the use of a prism, showed that sunlight can be decomposed into different colors. (Experiment on the dispersion of light)
- To describe this continuous array of colors produced by the prism, Newton coined the Latin word 'Spectrum'—meaning 'apparition' or 'ghost'—in his 1672 paper.



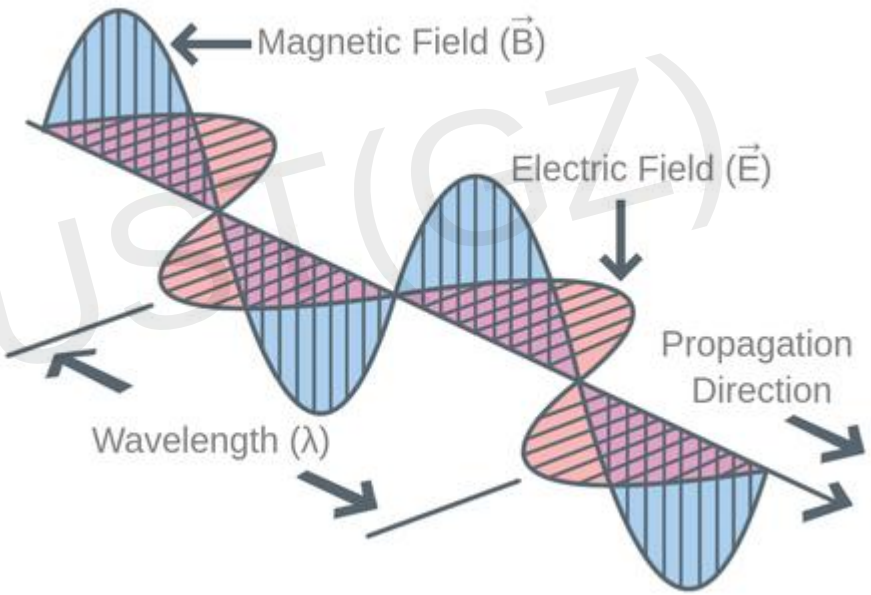


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What is Electromagnetic Wave?

- When charged particles (e.g. electrons) accelerate or decelerate, electromagnetic wave (EM) wave is generated and radiated at the speed of light, which is the highest speed anything can travel at:

$$c \approx 3 \times 10^8 \text{ m/s}$$



Electromagnetic waves | Physics | Khan Academy

- Unlike sound, EM wave can travel in vacuum.

- All EM waves have the same speed in a vacuum, but they can differ in wavelength

$$c = \frac{\lambda}{T} = \lambda \cdot f$$

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https://www.youtube.com/watch?v=k3GQYBD5hbI&t=913s

ORIGIN OF ELECTROMAGNETIC WAVES

James Clerk Maxwell

Electromagnetic Wave In Our Life

- The universe is filled with ***electromagnetic waves*** at a wide range of frequency spectrum.



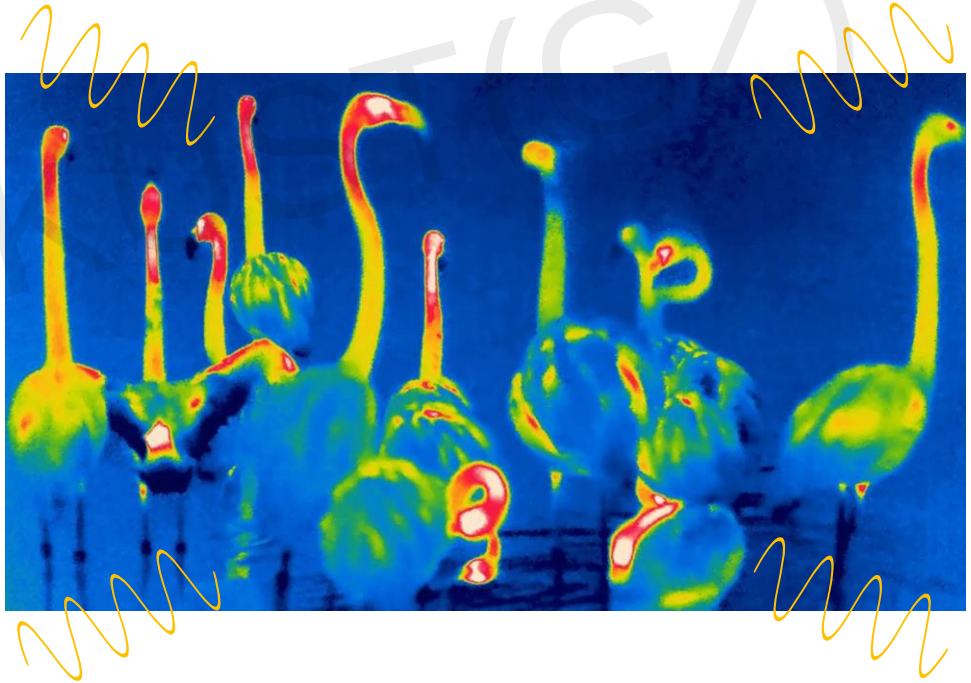


Figure source：https://www.coolaboo.com/astronomy/the-sun/， https://healthcare.utah.edu/healthfeed/2023/07/spot-signs-of-heat-exhaustion，https://www.eyemichigan.com/exploring-the-human-eye/, https://www.londonzoo.org/zoo-stories/news/thermal-images-animals-taken-overnight-london-zoo

Electromagnetic Wave In Our Life

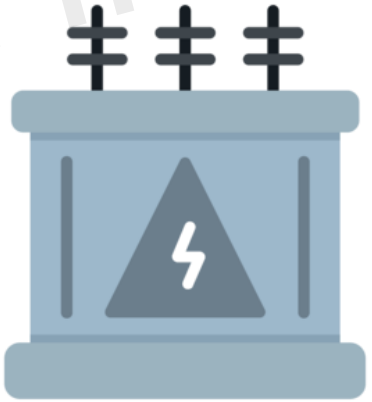
- The universe is filled with ***electromagnetic waves*** at a wide range of frequency spectrum.



Communication and Broadcasting Systems



Home Appliances



Electric Power and Industrial Equipment



Medical and Scientific Research Equipment

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Figure source: https://www.ylxyyy.com/article/1241119143825/ https://zh.pngtree.com/freepng/microwave-clipart-cartoon-pie-in-microwave-vector_11073628.html, https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRNgXSMF3JAjXCPH5eBINp_IAWlSfkly1qO8tx4R3Vd2A&s=10

Presence of Electromagnetic Wave

- How do we know of the presence of an EM wave? The radiated EM wave exerts a force on remote charged particles!
- When the oscillating electric field of a wave encounters a charged particle, it makes that particle oscillate in response.
  - This forced oscillation is the primary "footprint" of an EM wave. We then measure the effects of this oscillation.
- Example:

- **Radio:** The EM wave drives free electrons in an antenna to oscillate, producing an alternating electrical signal that can be detected and processed.
- **Microwave:** The oscillating electric field causes polar molecules, especially water molecules, to reorient repeatedly, converting electromagnetic energy into heat.
- **Human eye**: Visible light is absorbed by light-sensitive molecules in the retina, causing molecular changes that eventually produce nerve signals and visual perception

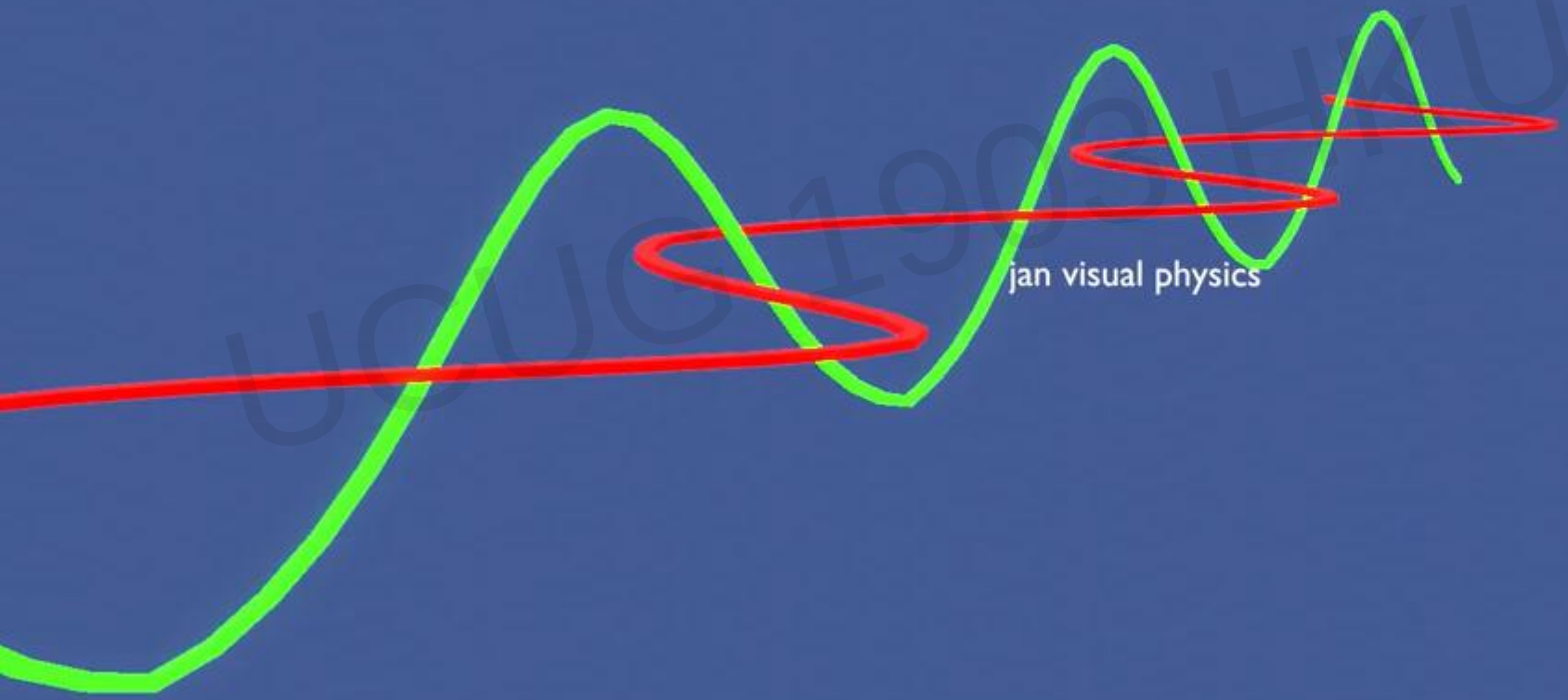
Radio Receiver Working Animation | How Radio Works | How Antenna Works | Antenna

An easy to understand explanation of how microwaves work

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https://www.youtube.com/watch?v=Aj1FuYjhTUQ

Radio wave Receiver



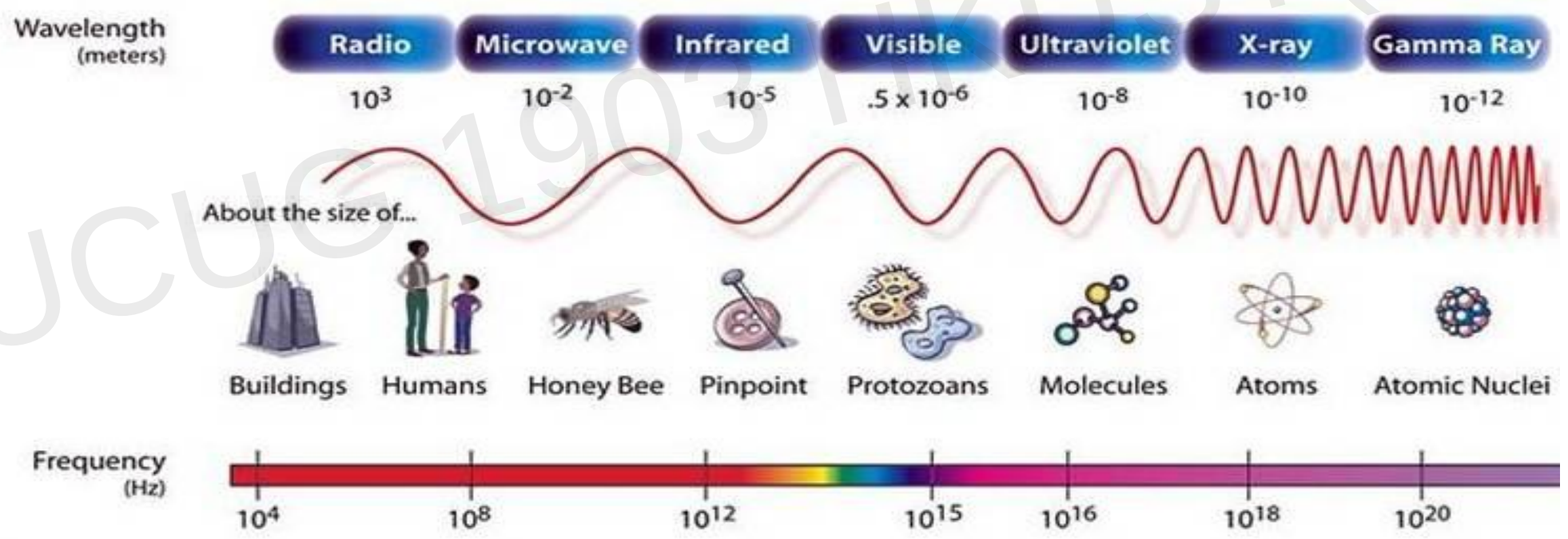
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https://www.youtube.com/watch?v=xDM_Gkpplck

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Wavelength for Applications

- Electromagnetic waves cover a very wide range of **wavelengths** and **frequencies**.
- Different wavelength ranges interact with matter in **different ways**.
- As a result, different parts of the electromagnetic spectrum are used for **different applications**.
- In general, the application of an EM wave is closely related to its **wavelength** (or equivalently, its frequency).



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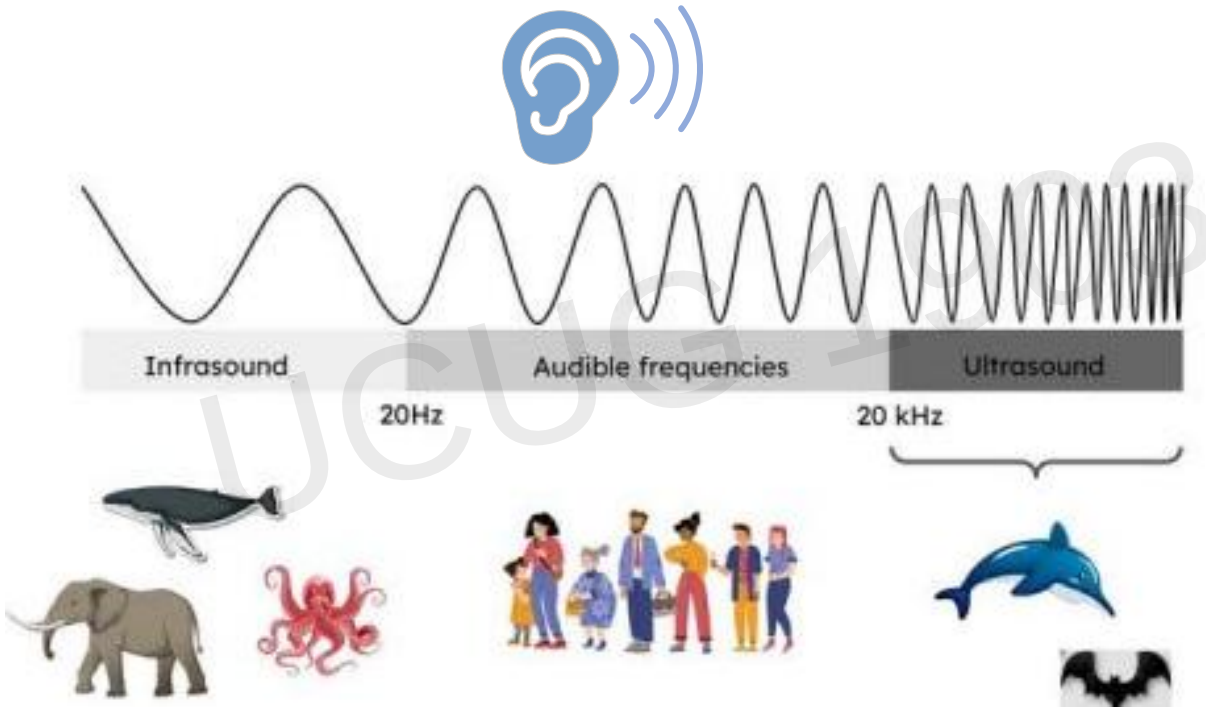
Source: ChatGPT

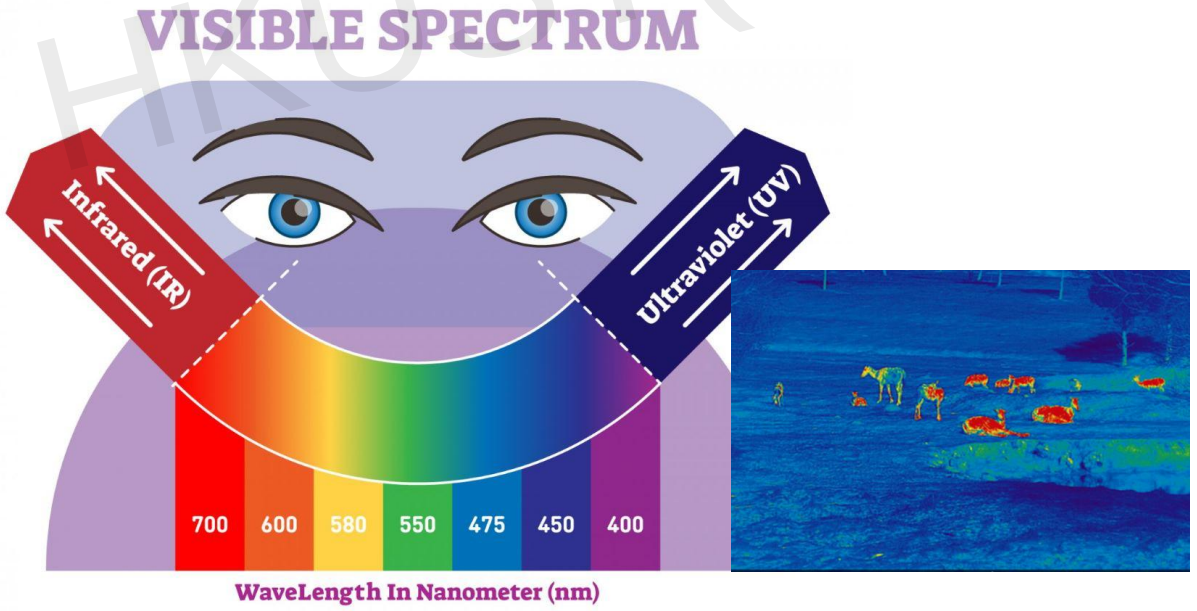
Filter

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Systems and Bandwidth

- As explained earlier, anything that processes signals can be modeled as a ***system.*** Both signals and systems can be thought of in the frequency domain.
- Physical systems inherently respond only to frequencies within specific ranges, ***working bandwidth*** of the system.

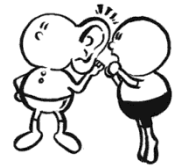




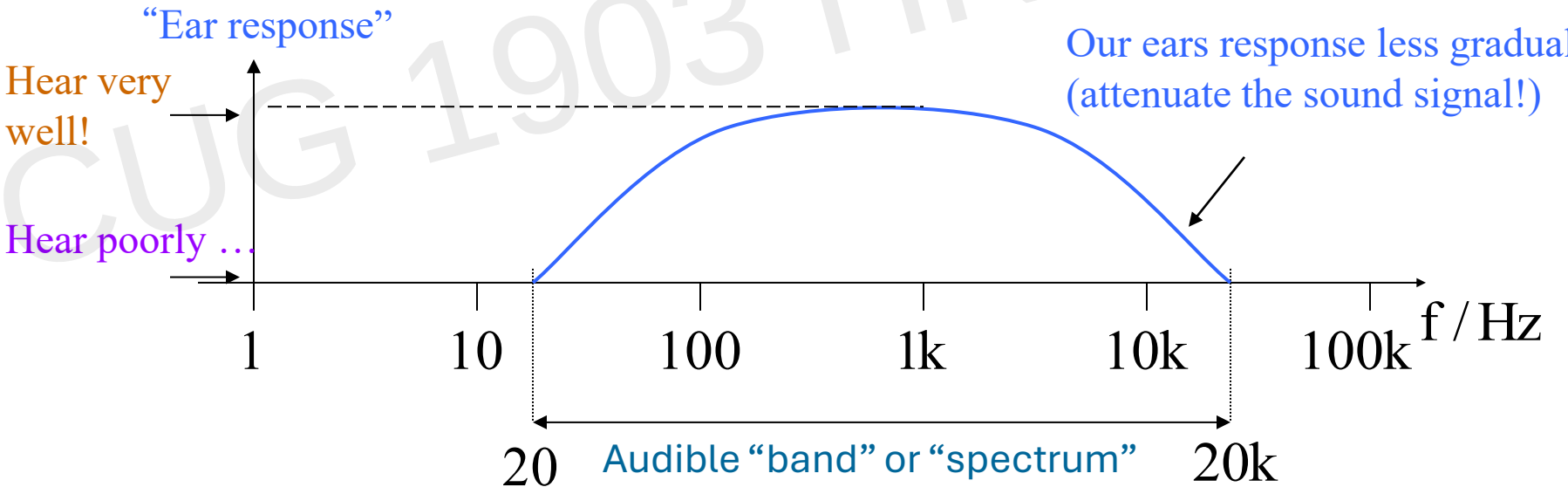
https://www.linkedin.com/posts/waleed-ahmed678_types-of-sound-by-frequency-some-interesting-share-7371861392204664833-4K3U/

https://www.mouqy.com/blog/visible-spectrum/?srsltid=AfmBOoqKto7d0ijb_u3YtP9xBDdpTldMW05V1IvrA5We86BhgWuqNeE5

Spectral Response of Human Ears



- For the same *physical level*, our ears hear the frequency range of 1 kHz to 5 kHz better than other frequencies.
- For *lower* and *higher* frequencies, e.g., below 60 Hz and above 10 kHz, our ears attenuate the signal and they sound weaker!
- In fact, we can describe the sensitivity of our ear to different frequencies by a typical "spectral response curve"



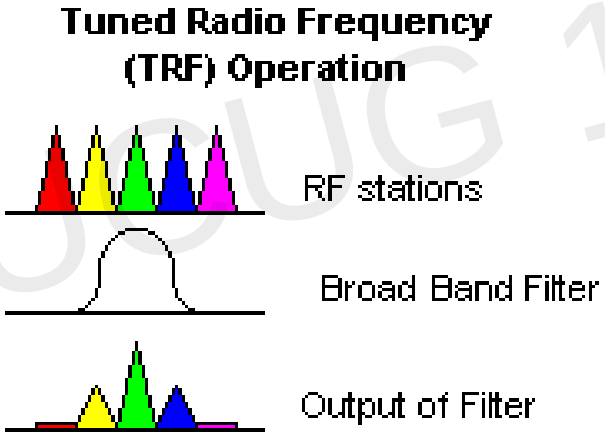
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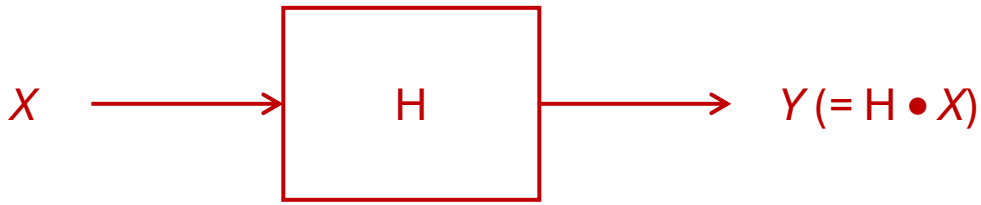
Signal Processing In The Frequency Domain

- We can conveniently view systems as ***filters*** in the frequency domain with a spectral response curve **H(f)**, so that if spectrum of the input is ***X(f)***, the *spectrum* of the output ***Y(f)*** would be the *product* of **H(f)** and ***X(f)***. (**H**, ***X***, ***Y*** are all functions of frequency)

input spectrum • filter spectral response = output spectrum

or $X \bullet H = Y$





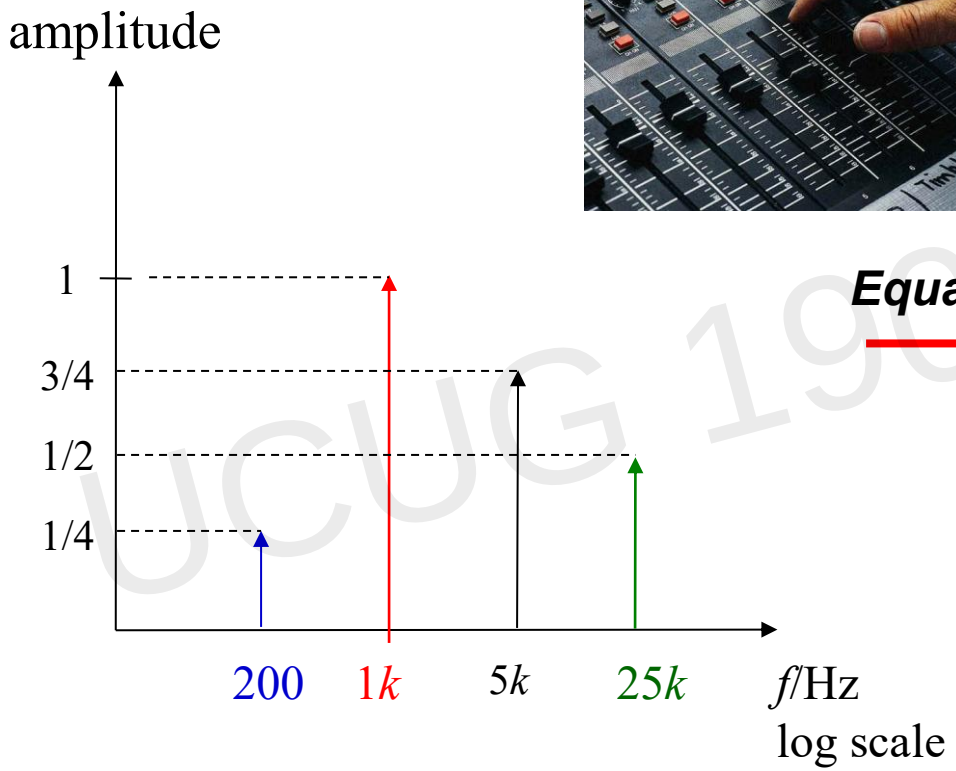
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Example: Equalizer

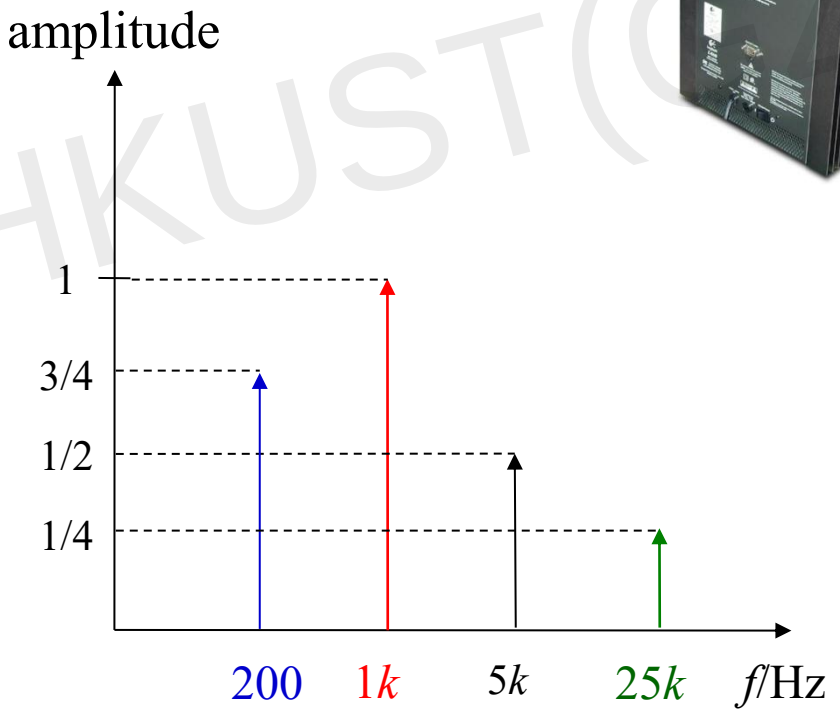
- In this example, the equalizer boosts up the amplitude of the low frequency (200 Hz) and attenuated the higher frequency (5 *k*Hz)







Equalize



https://www.youtube.com/watch?v=vRa_Zk90lDc

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Example: Equalizer

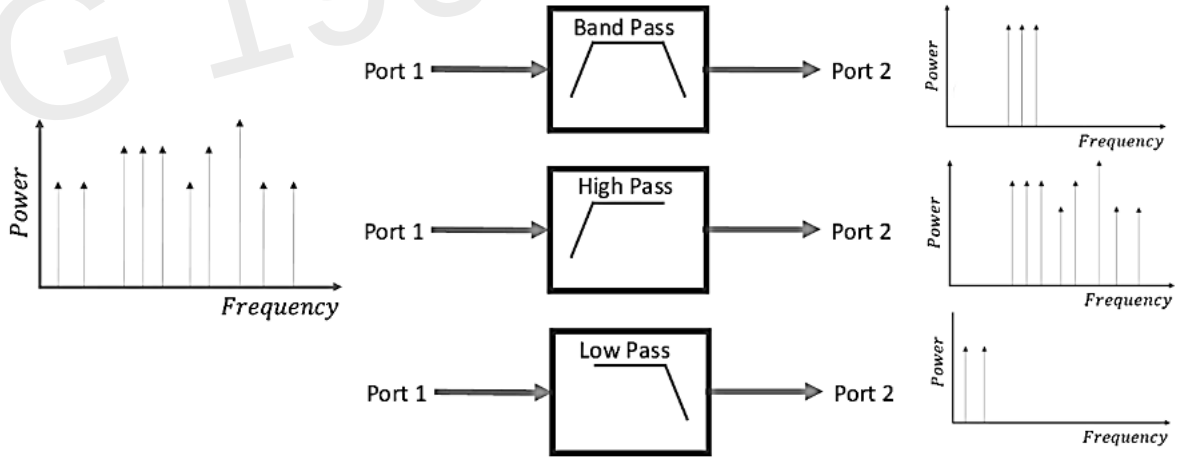


https://www.youtube.com/watch?v=vRa_Zk90lDc

Lowpass, Bandpass and Highpass Filters

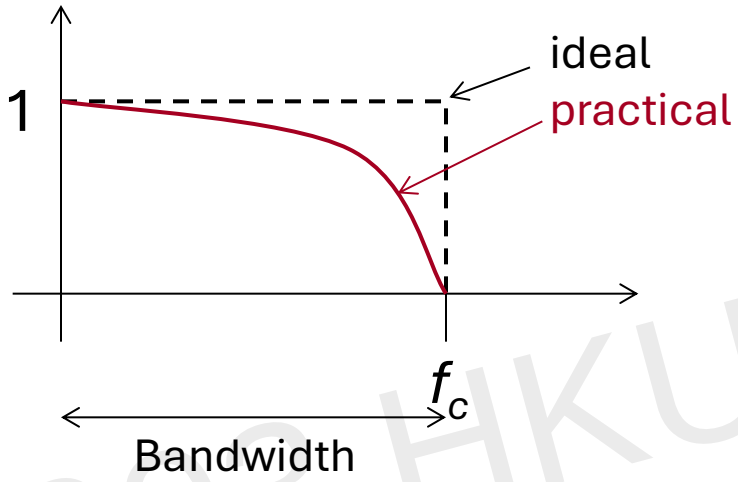
- Engineers frequently build three types of generic filters – the ***lowpass***, ***bandpass*** and ***highpass*** filters. They are the basic building blocks of all signal processing systems. They can be implemented as electrical circuits or in computer programs.

- They are defined in **ideal forms**, but there are fundamental reasons that the ideal form *cannot be perfectly realized*, neither by circuits nor by computer programs. In real systems, we build **practical filters** that are as close to the ideal form as possible subject to different criteria.



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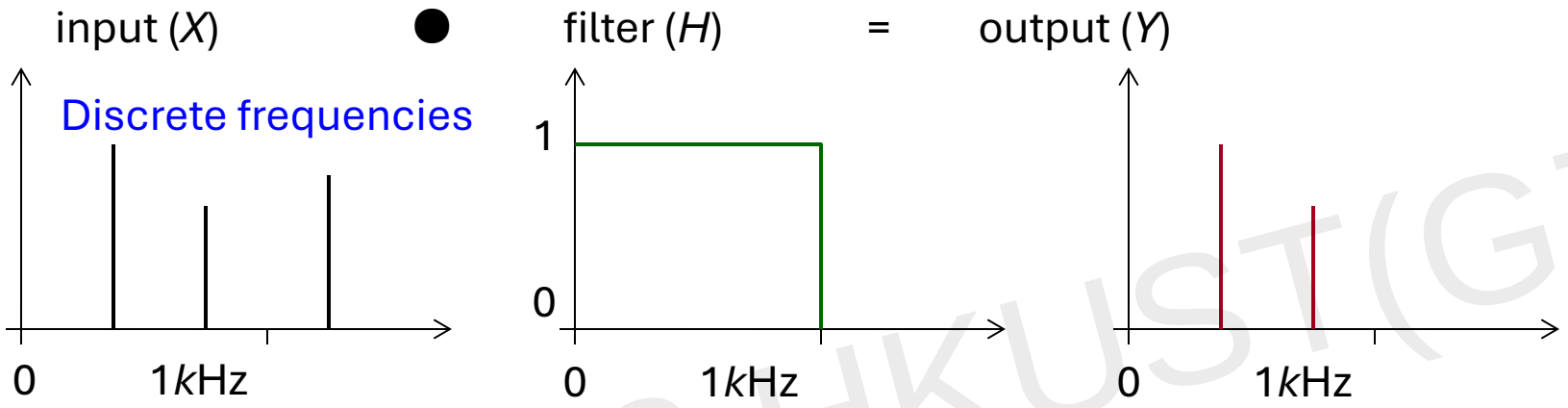
Lowpass Filters

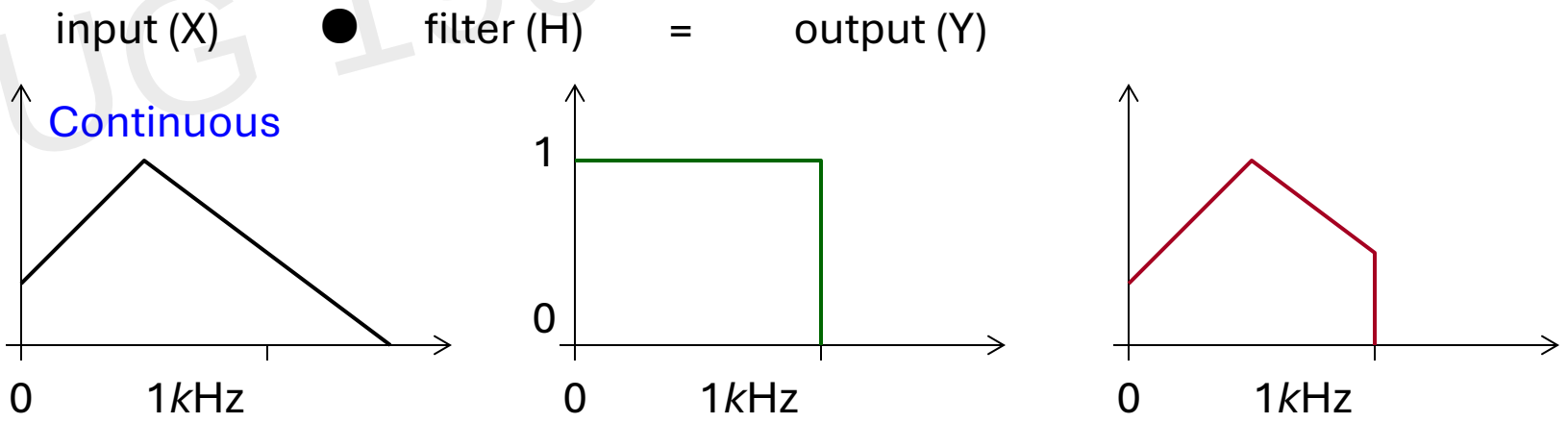


- A **lowpass filter** only allows signal with low frequencies to pass through.
- f_c is the designed "**cutoff frequency.**" Sine waves with $f < f_c$ can pass through. Sine waves with $f > f_c$ are blocked.
- The **filter bandwidth** is f_c.

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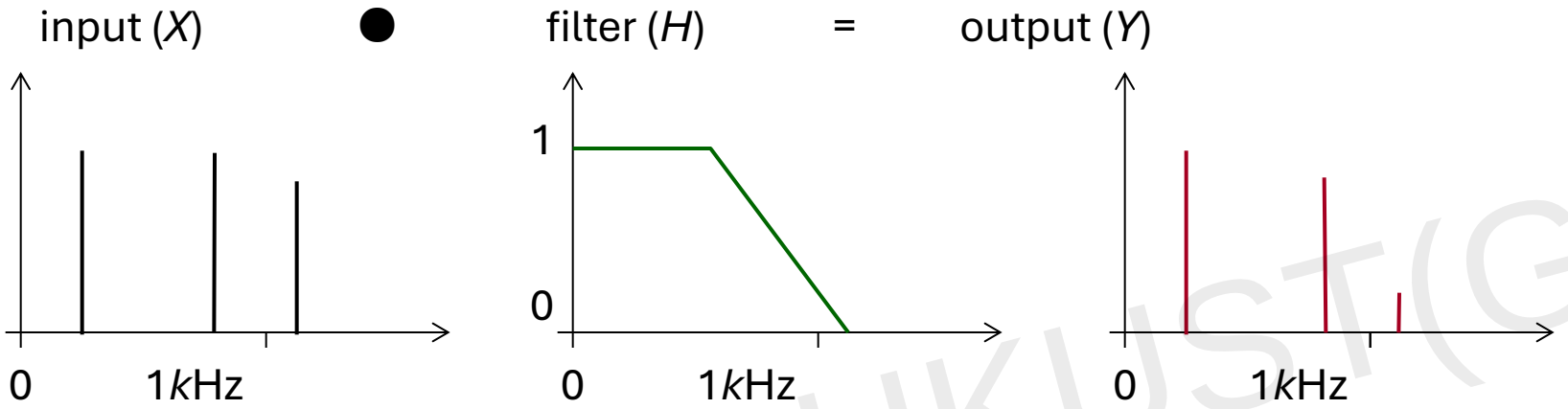
Operations with an Ideal Lowpass Filter

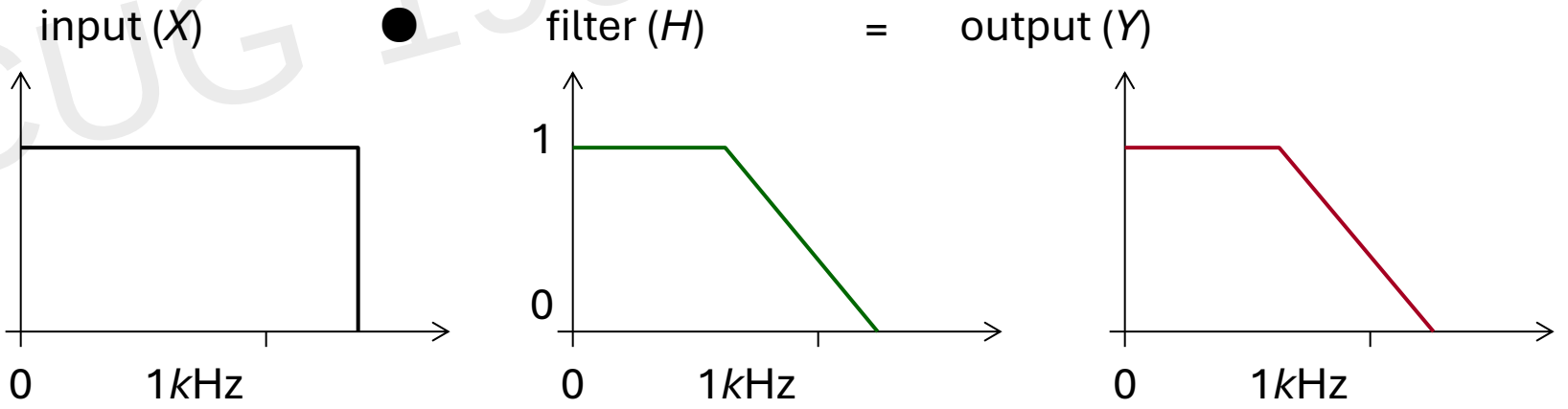




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Operations with a Practical Lowpass Filter

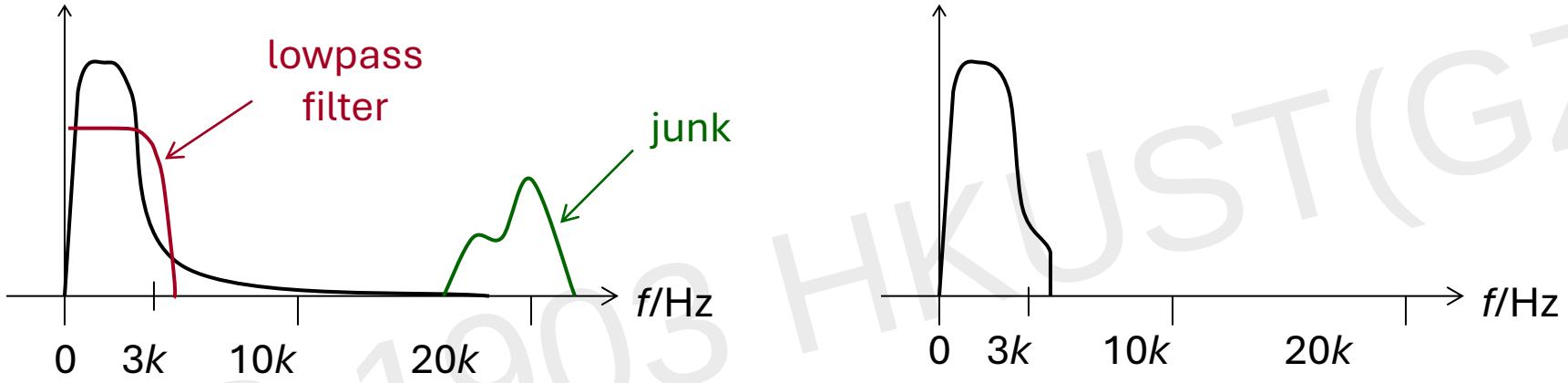




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Lowpass Filtering of Telephone Voice Spectrum

- The traditional telephone network often uses a lowpass filter with cutoff frequency of ~3 *k*Hz to filter the voice signal coming in from the telephone line.



- By lowpass filtering the signal, we suffer some distortion but as the voice signal has little energy at high frequency, the distortion is tolerable. In exchange, we have less costly communications equipment which can now universally assume that the maximum voice signal frequency is 4 *k*Hz only. We also prevented junk signals at high frequencies from getting into the network.

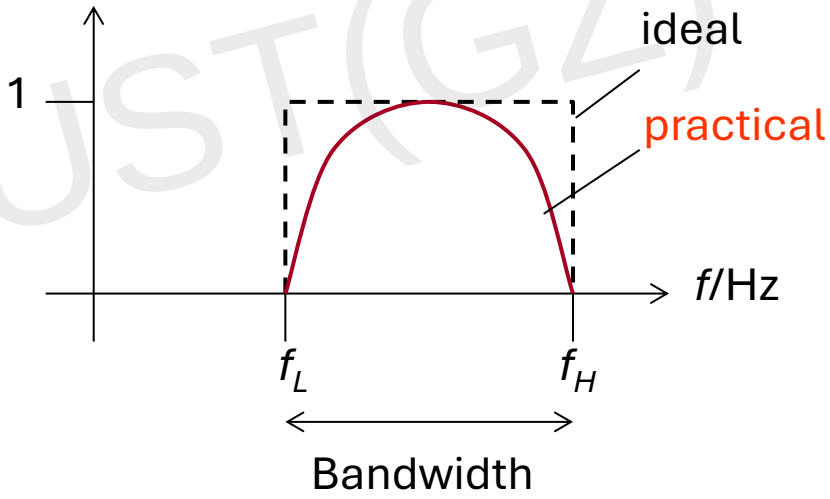
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Bandpass Filters

- A ***bandpass*** filter only allows signals within the designed frequency range to pass through.

- f_L and f_H are **lower** and **upper cutoff frequencies**. Signals with $f_H > f > f_L$ can pass through. Other frequencies are blocked.

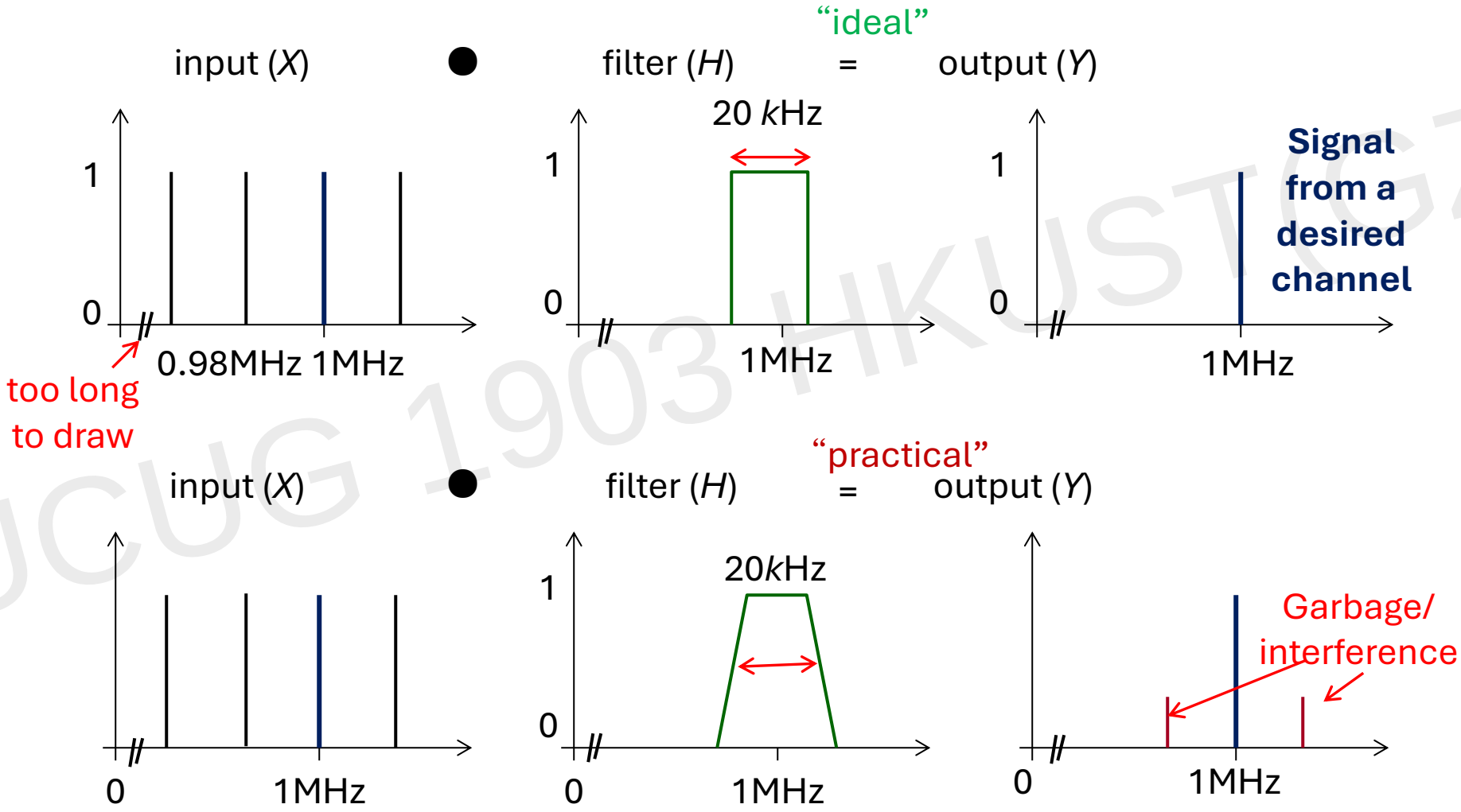
- $f_H - f_L$ is the **filter bandwidth** (e.g. *ears as bandpass filters have a typical bandwidth of from 20 Hz to 20 KHz; tuner for an analog TV has a filter bandwidth of 6 MHz to extract the audio and video signal from the desired radio channel*)



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Selecting Channels with Bandpass Filters

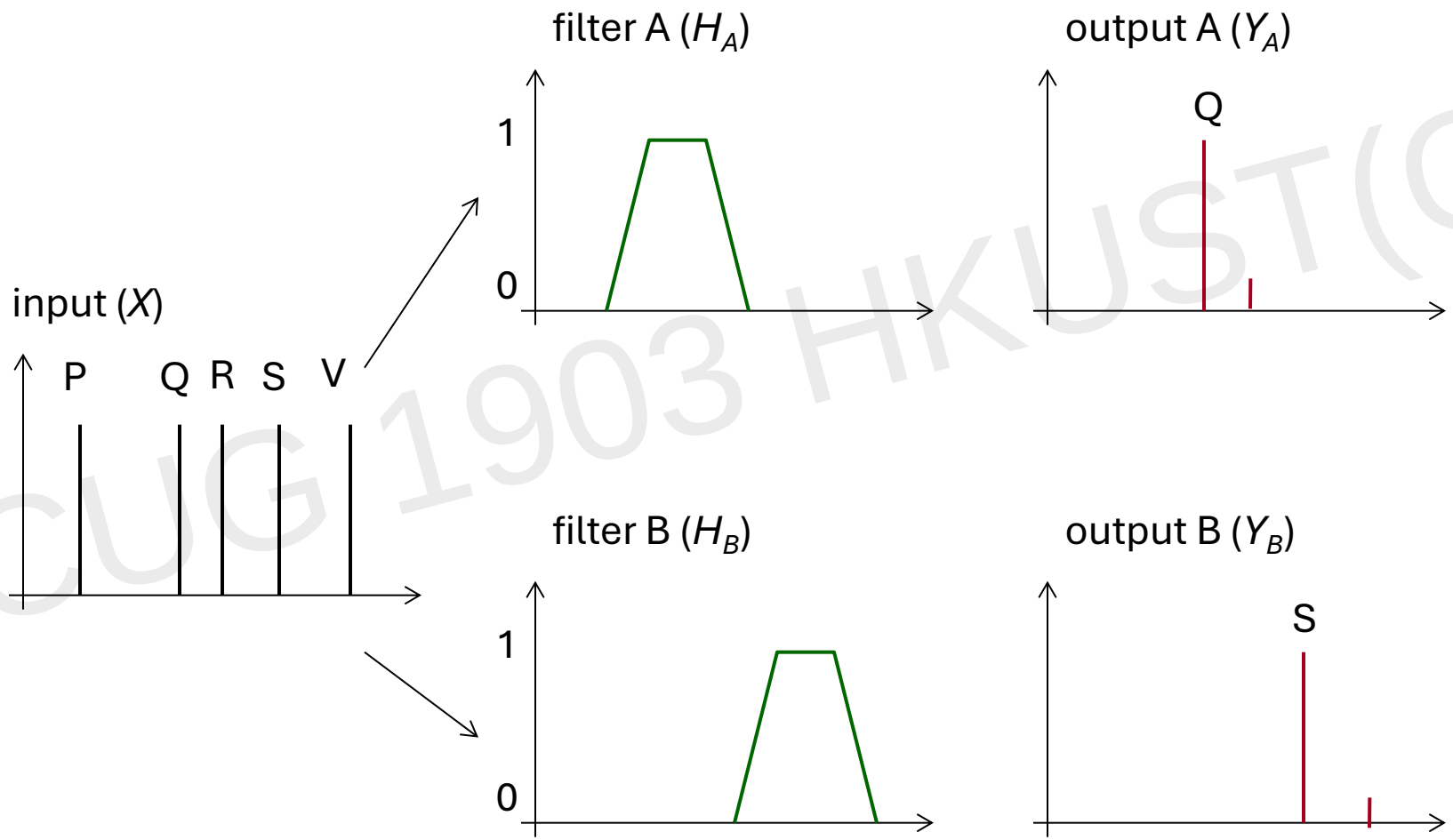
- e.g. In a radio, we select a channel at a particular frequency band by using a (bandpass) filter.



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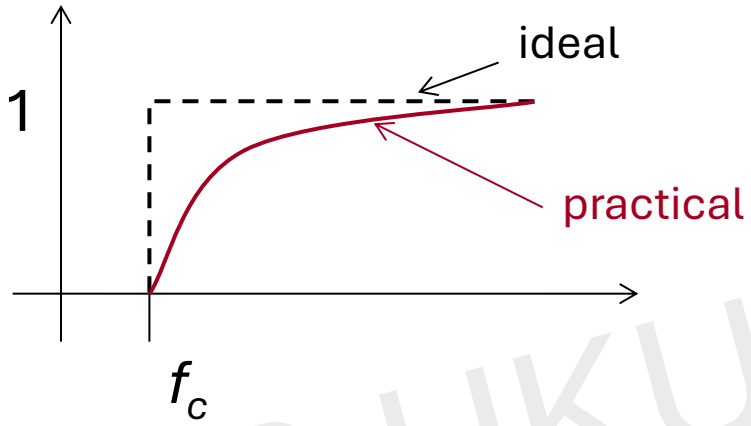
Different Bandpass Filters Select Different Channels

- By tuning the passband of the filter, we can select different radio signals



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Highpass Filters

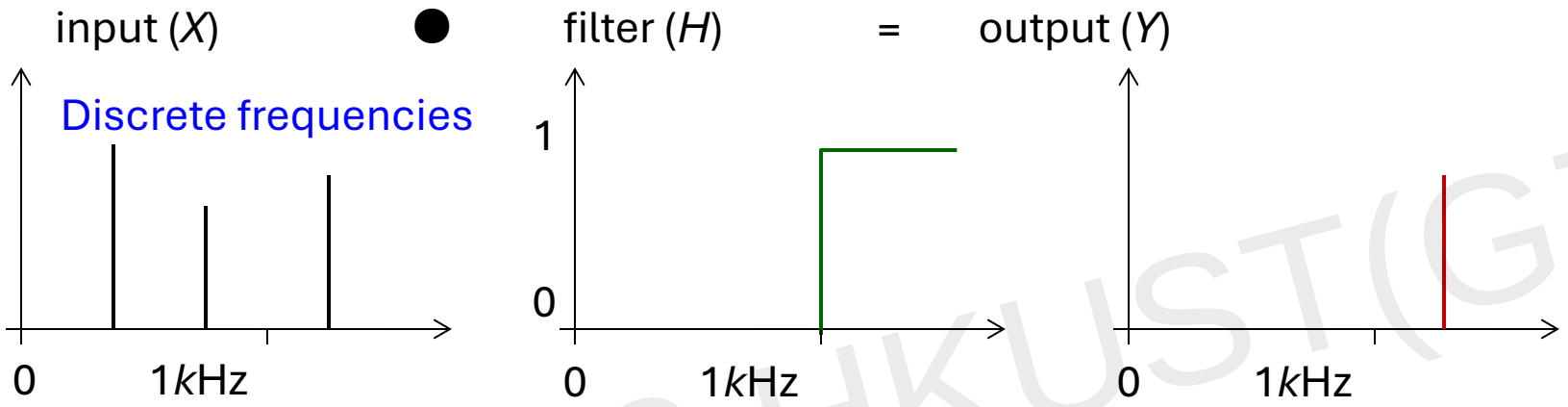


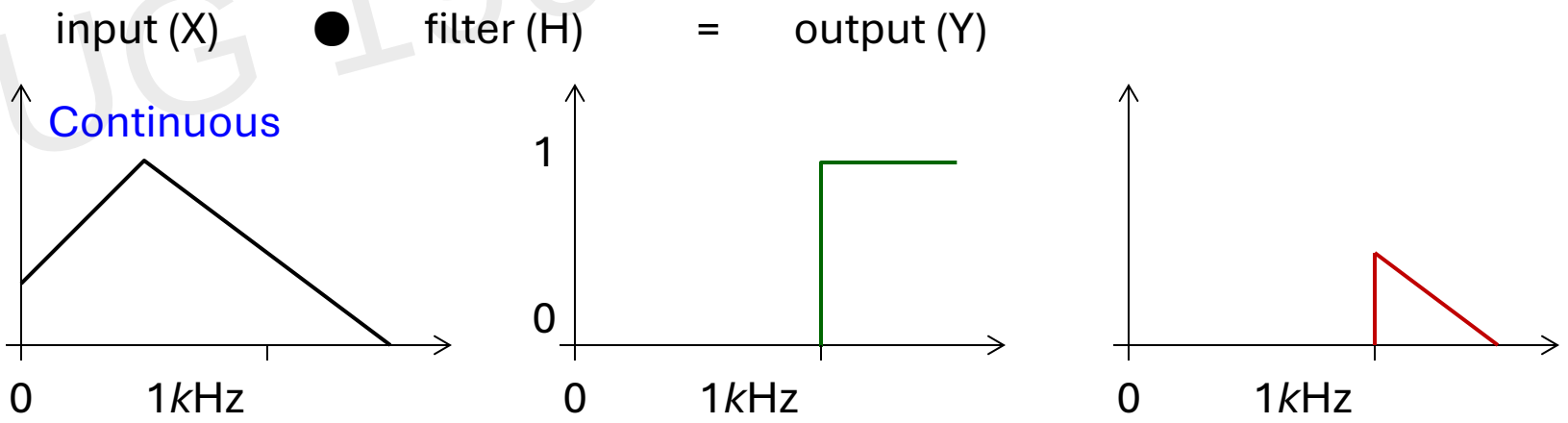
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- A **highpass filter** only allows signal with high frequencies to pass through.

- f_c is the designed "**cutoff frequency.**" Sine waves with $f > f_c$ can pass through. Sine waves with $f < f_c$ are blocked.

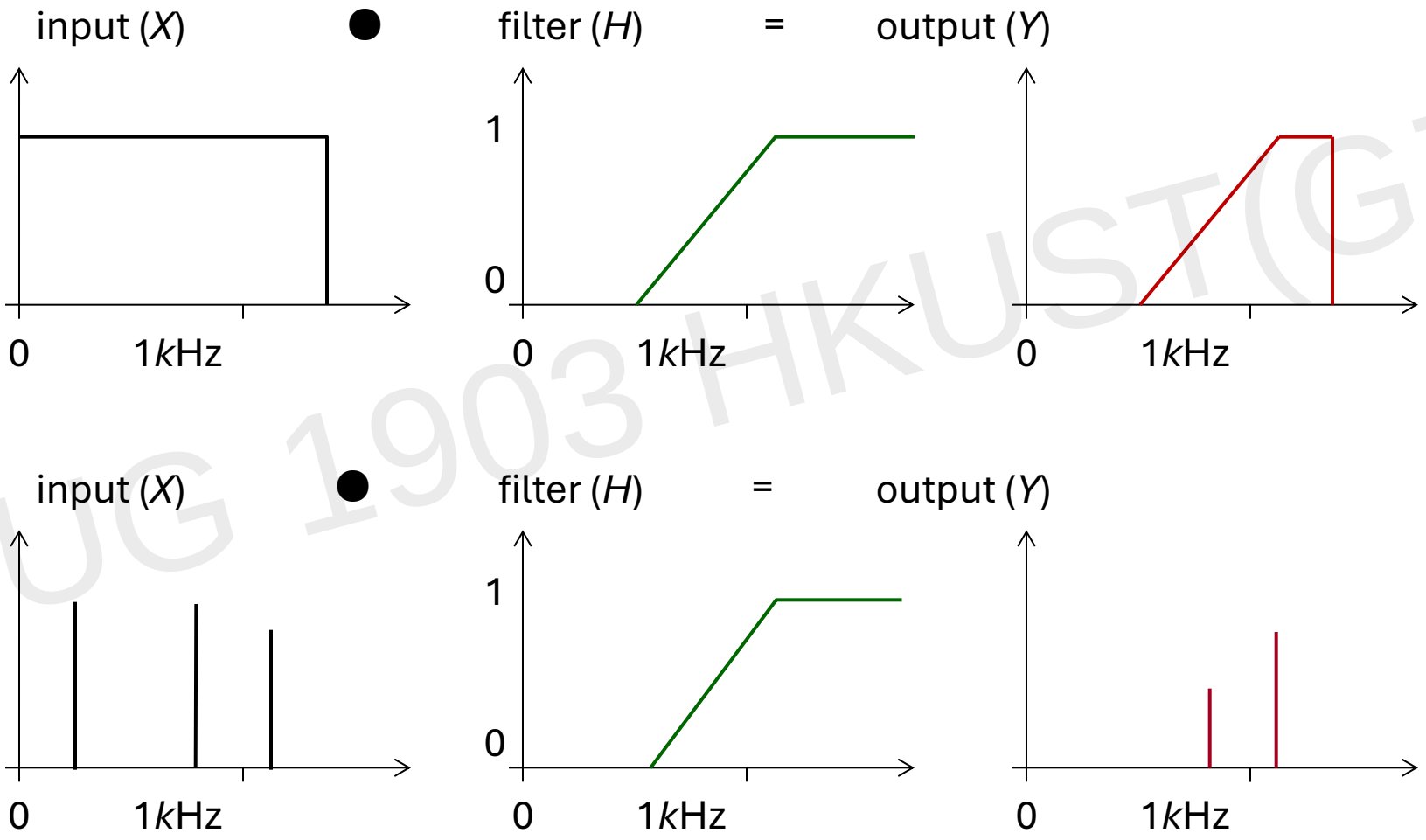
Operations with An Ideal Highpass Filter





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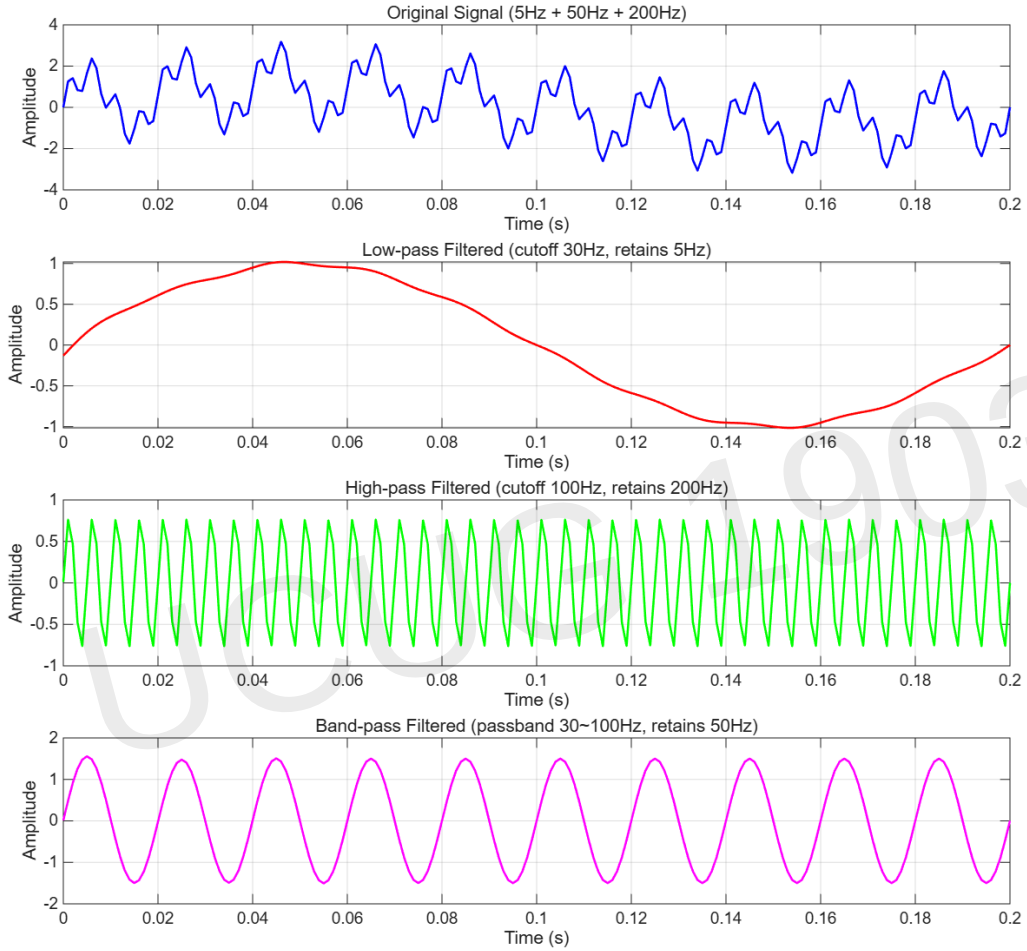
Operations with A Practical Lowpass Filter

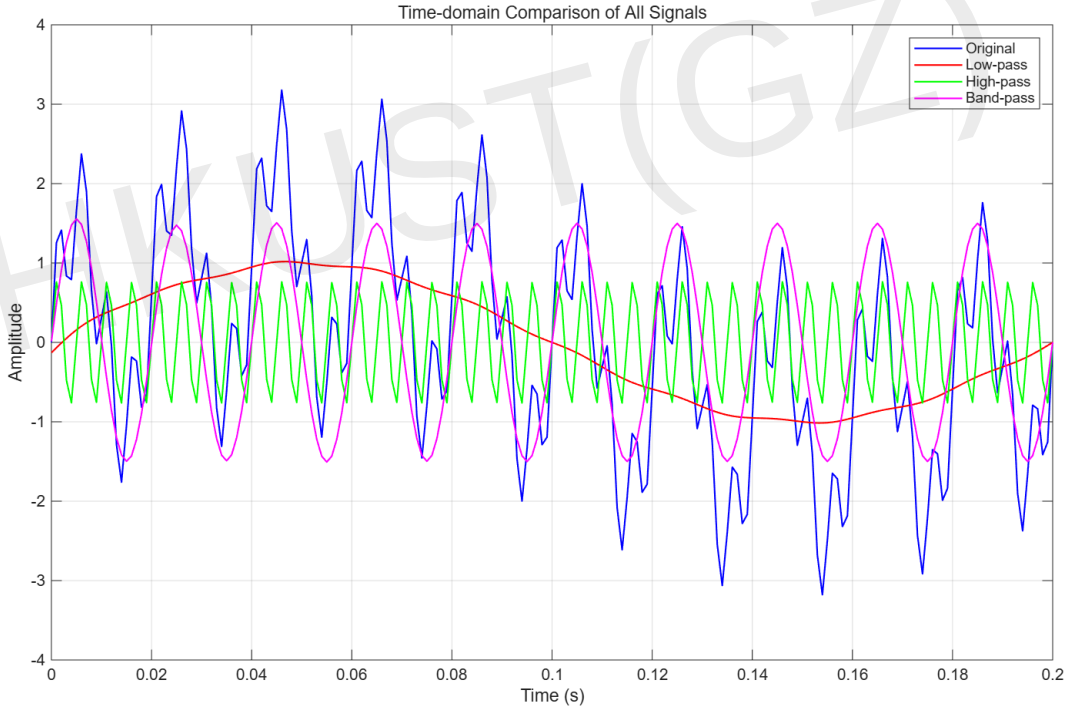


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Filters

- The impact on time domain after applying filters





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Signal Processing Demo

- In this demo, we can see there are many simple but useful operations using signal processing techniques in the frequency domain:
  - Lowpass Filtering
  - Bandpass Filtering
  - Highpass Filtering

- Real life application
  - Audio denoising
  - Beautification



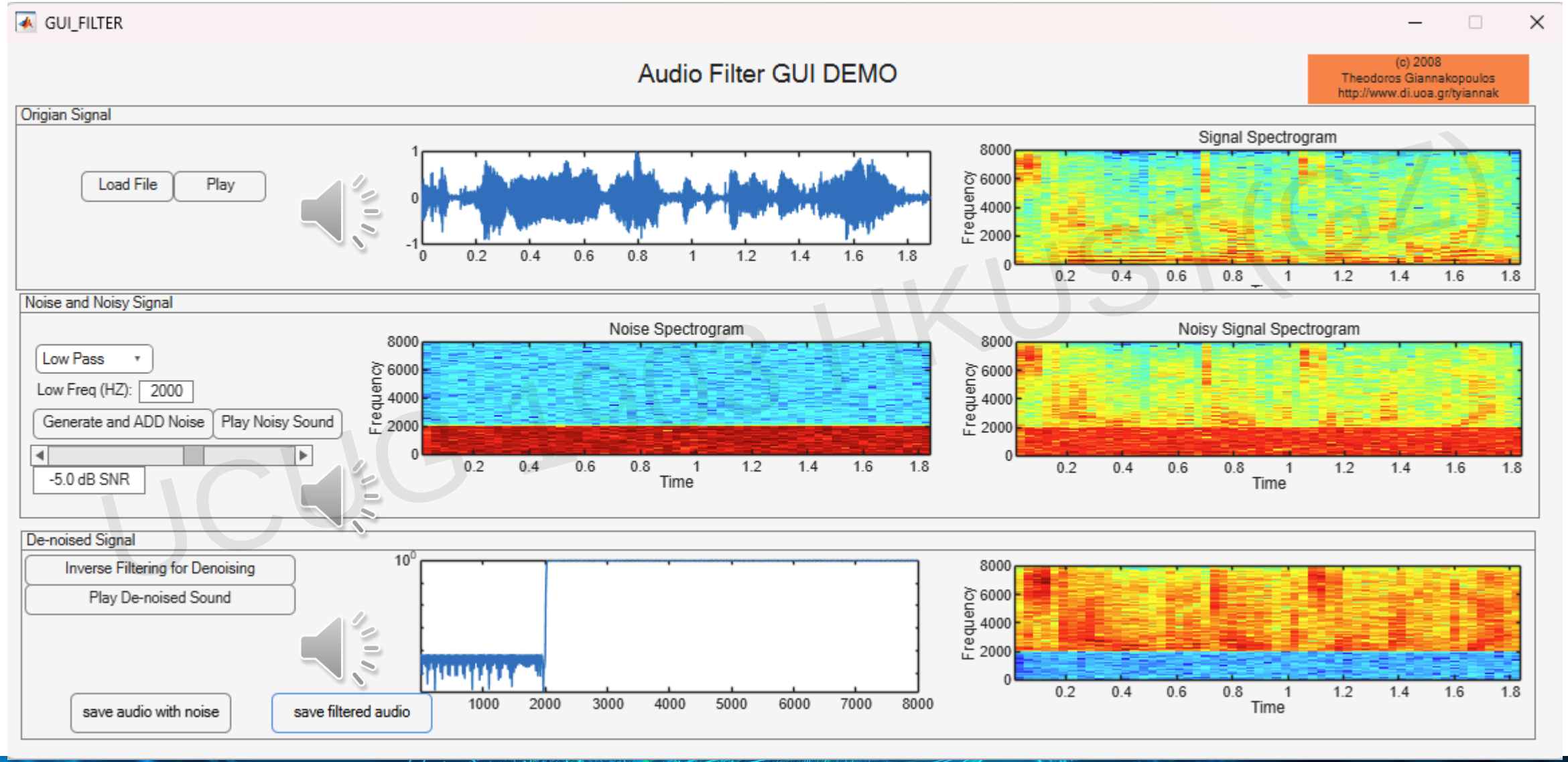




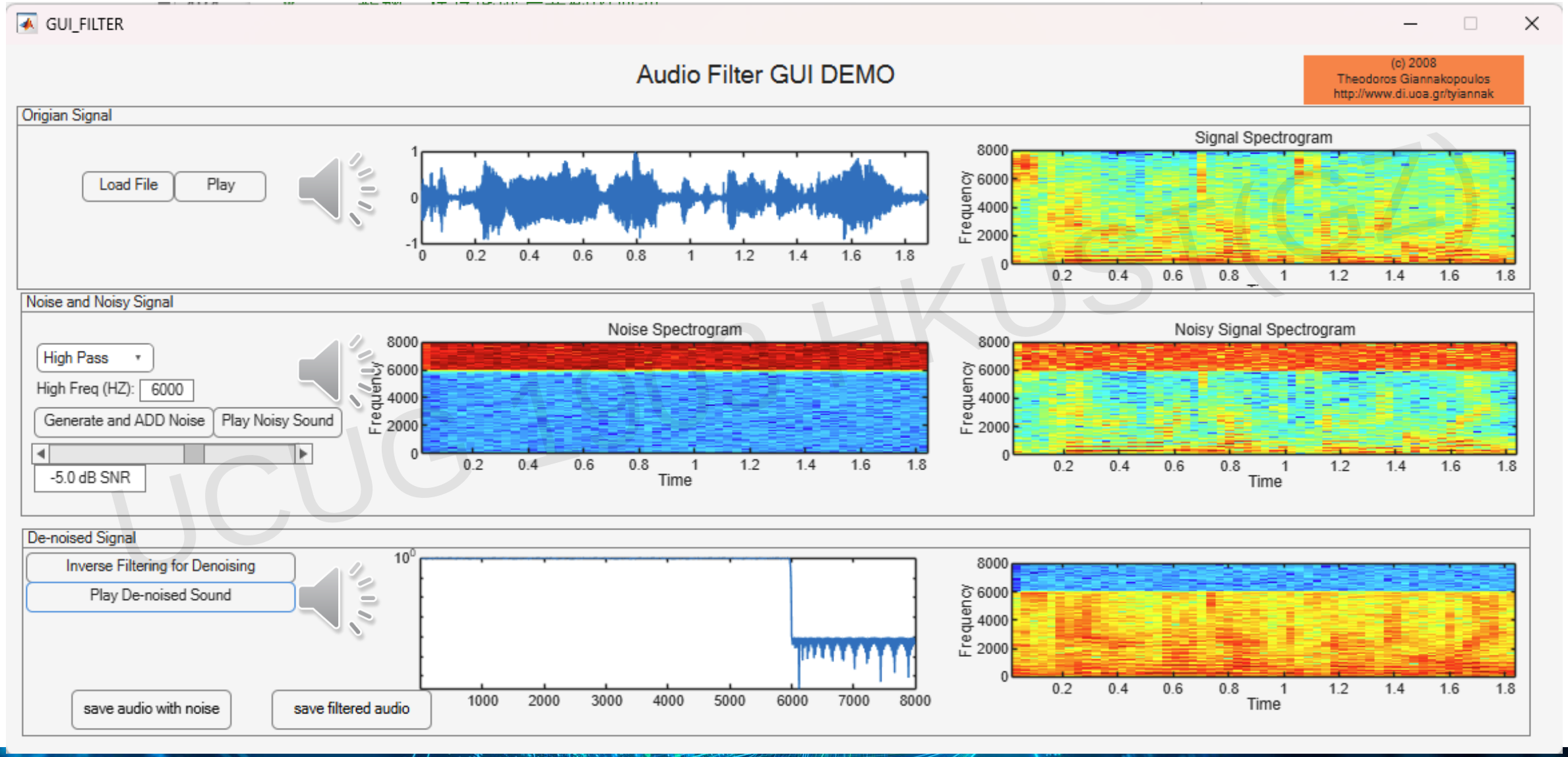


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Audio Denosing

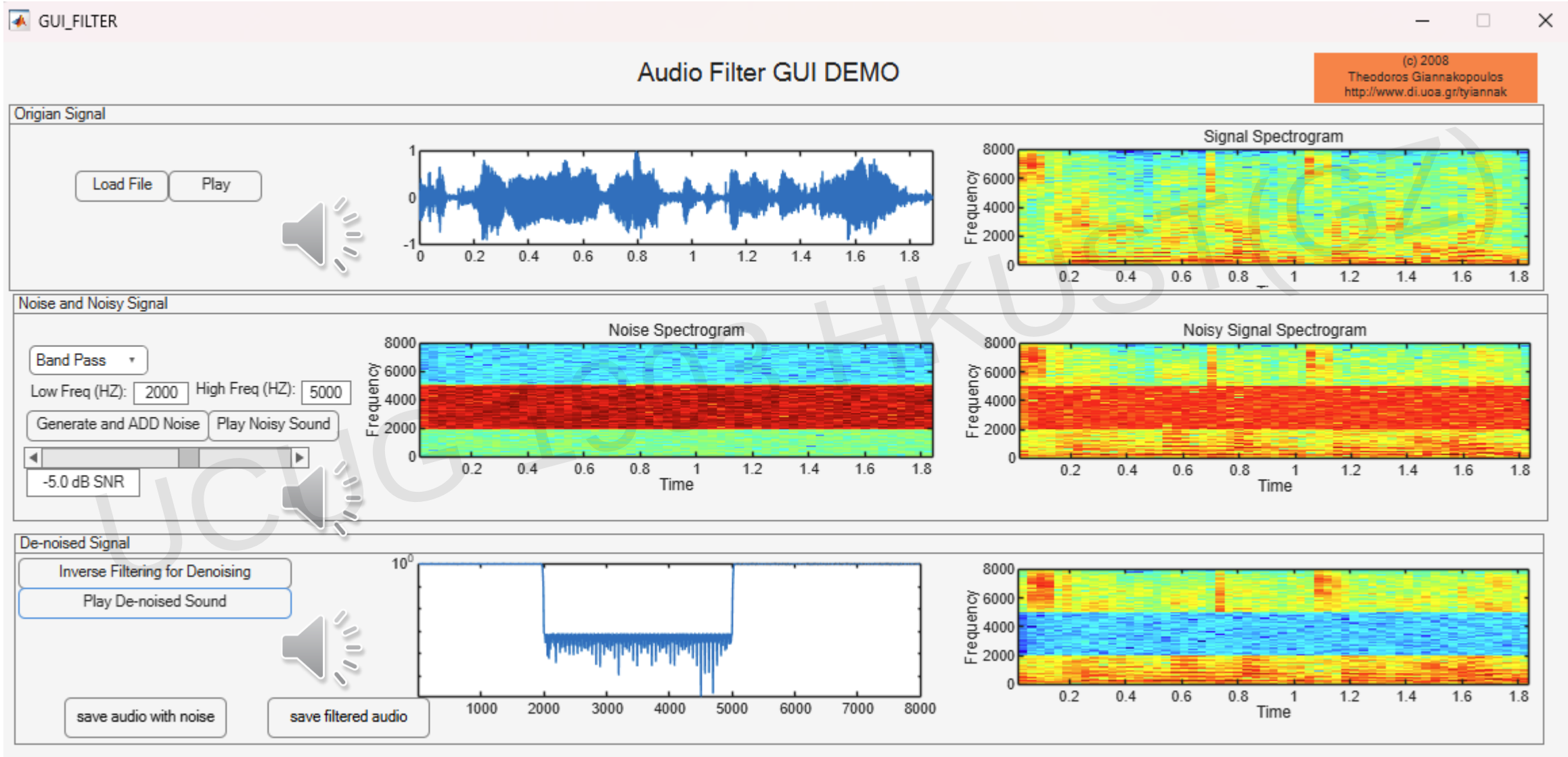


Audio Denosing



48

Audio Denosing



49

Beautification

Original



Bilateral Filter (smooth=160.0)

